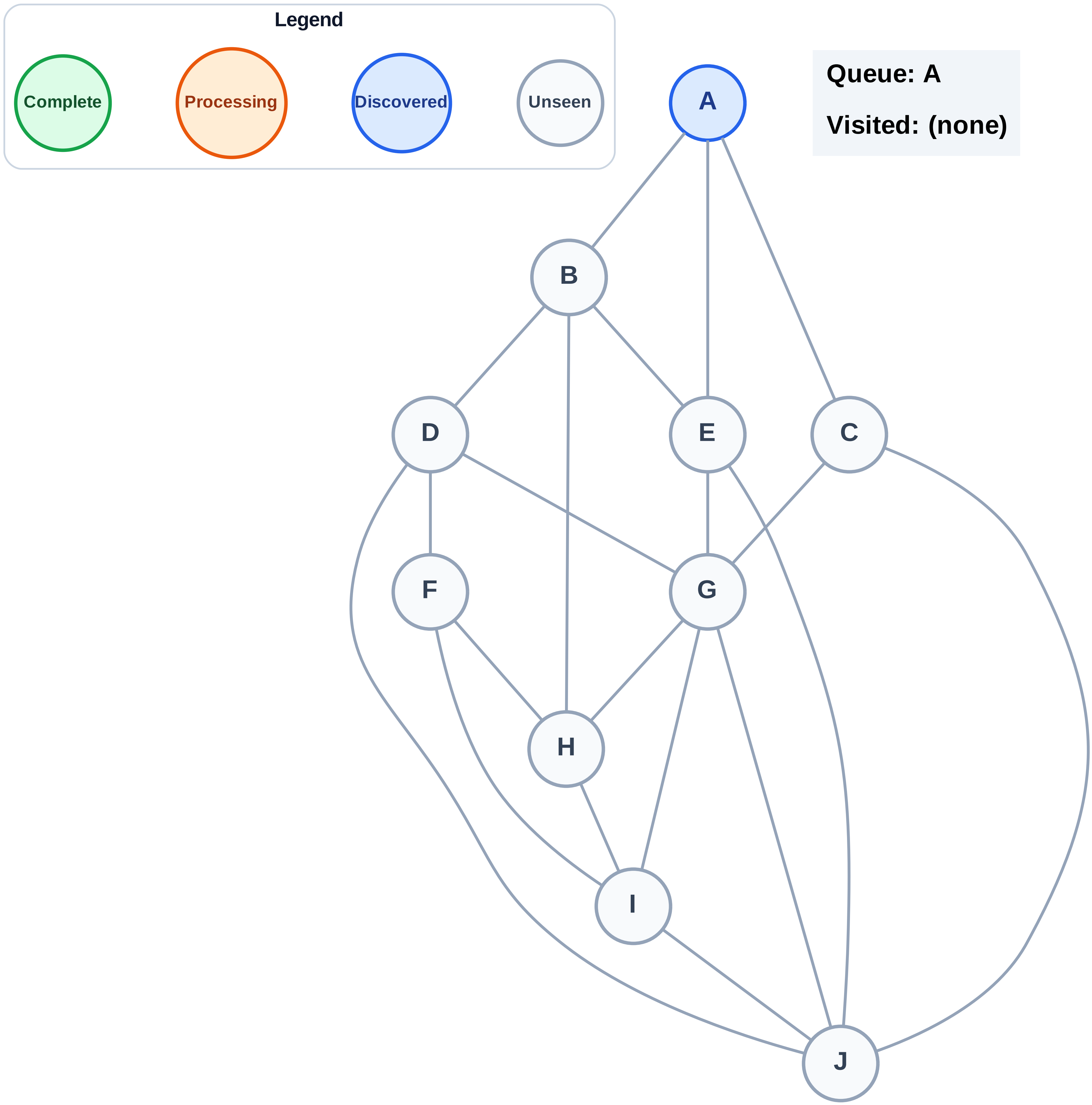
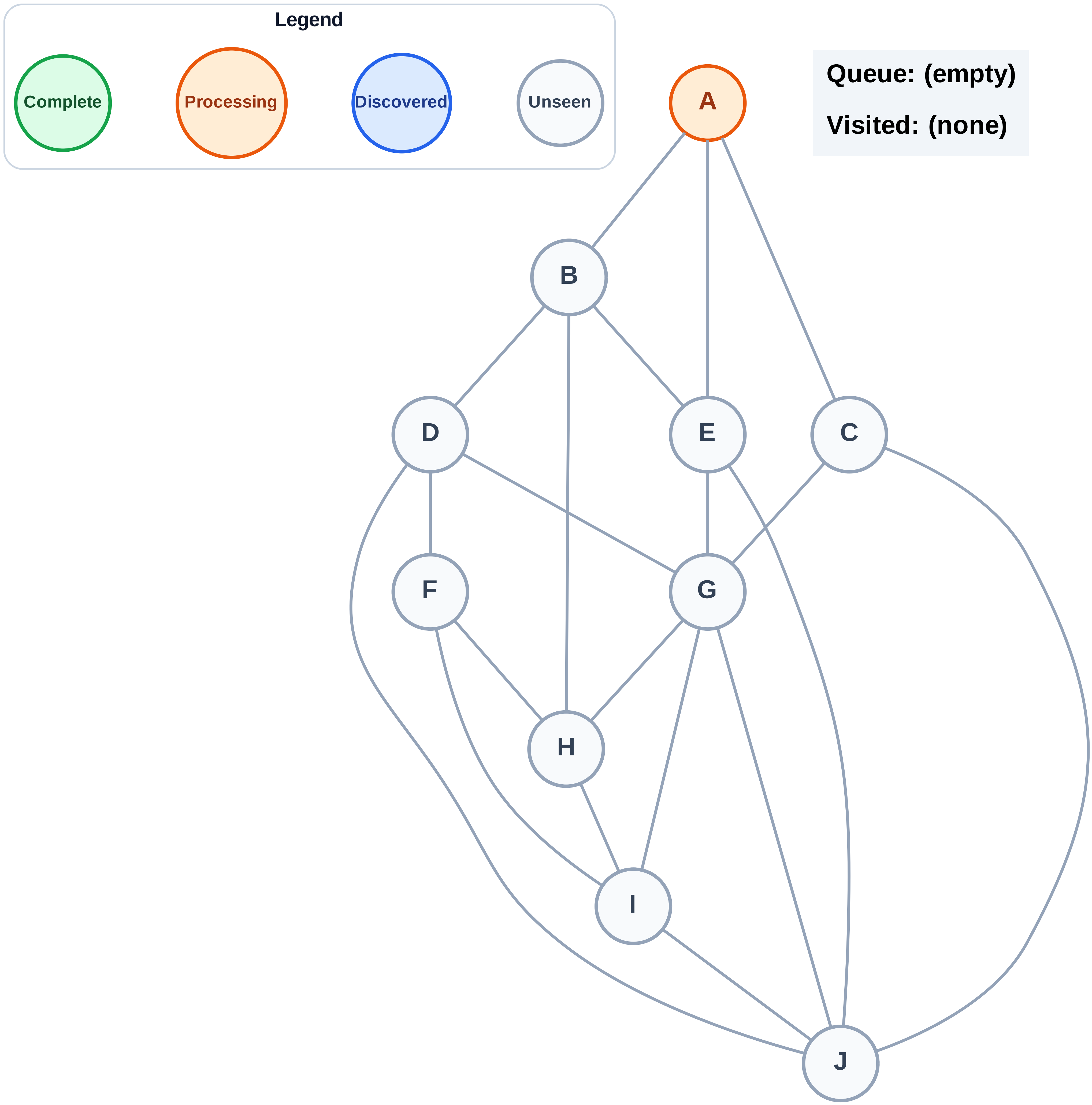
BFS Traversal

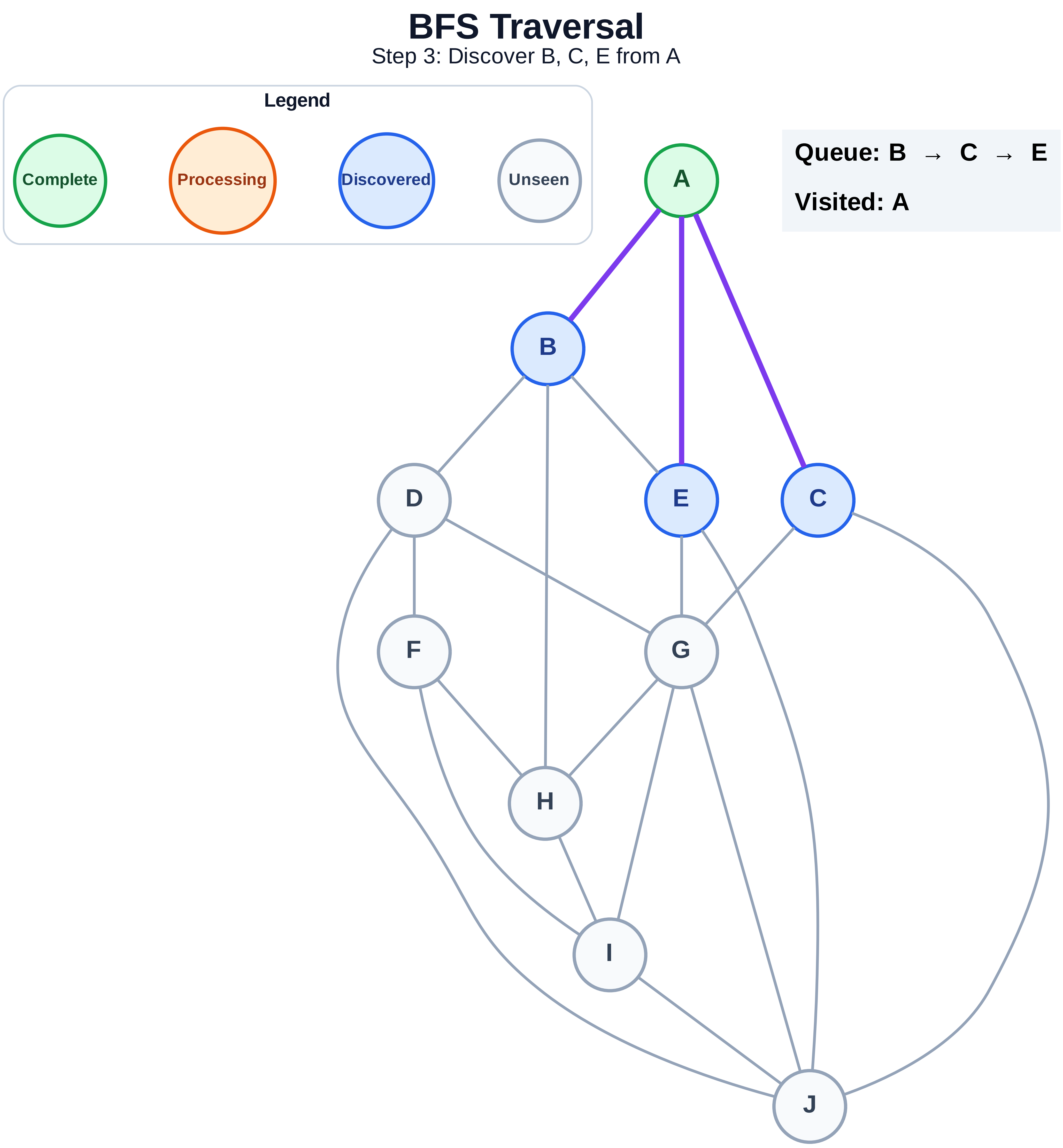
Step 1: Start at A



BFS Traversal

Step 2: Process node A

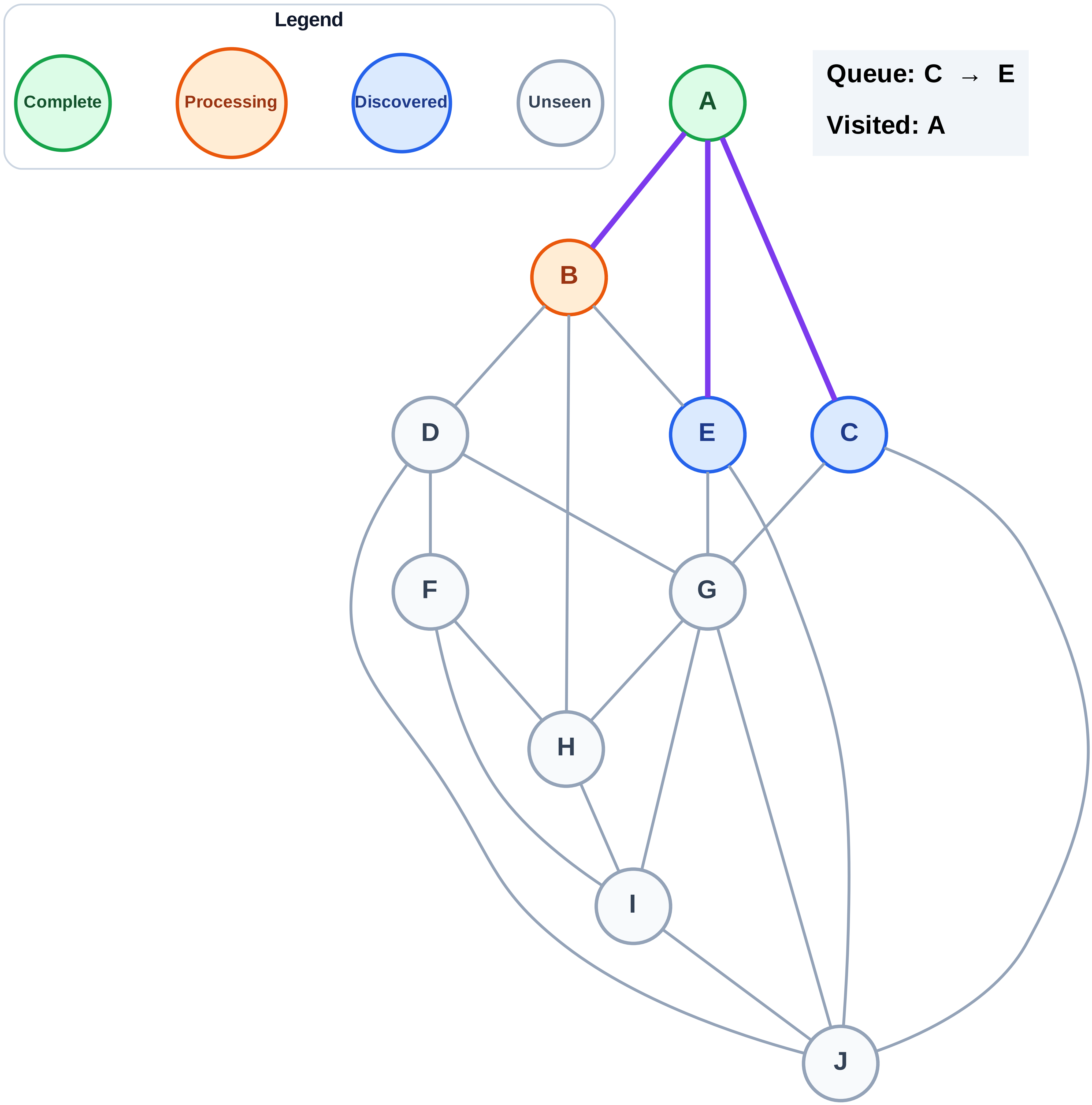




BFS Traversal
Step 3: Discover B, C, E from A
Legend
Complete
Processing
Discovered
Unseen
Queue: B → C → E
Visited: A
A
B
D
E
C
F
G
H
I
J

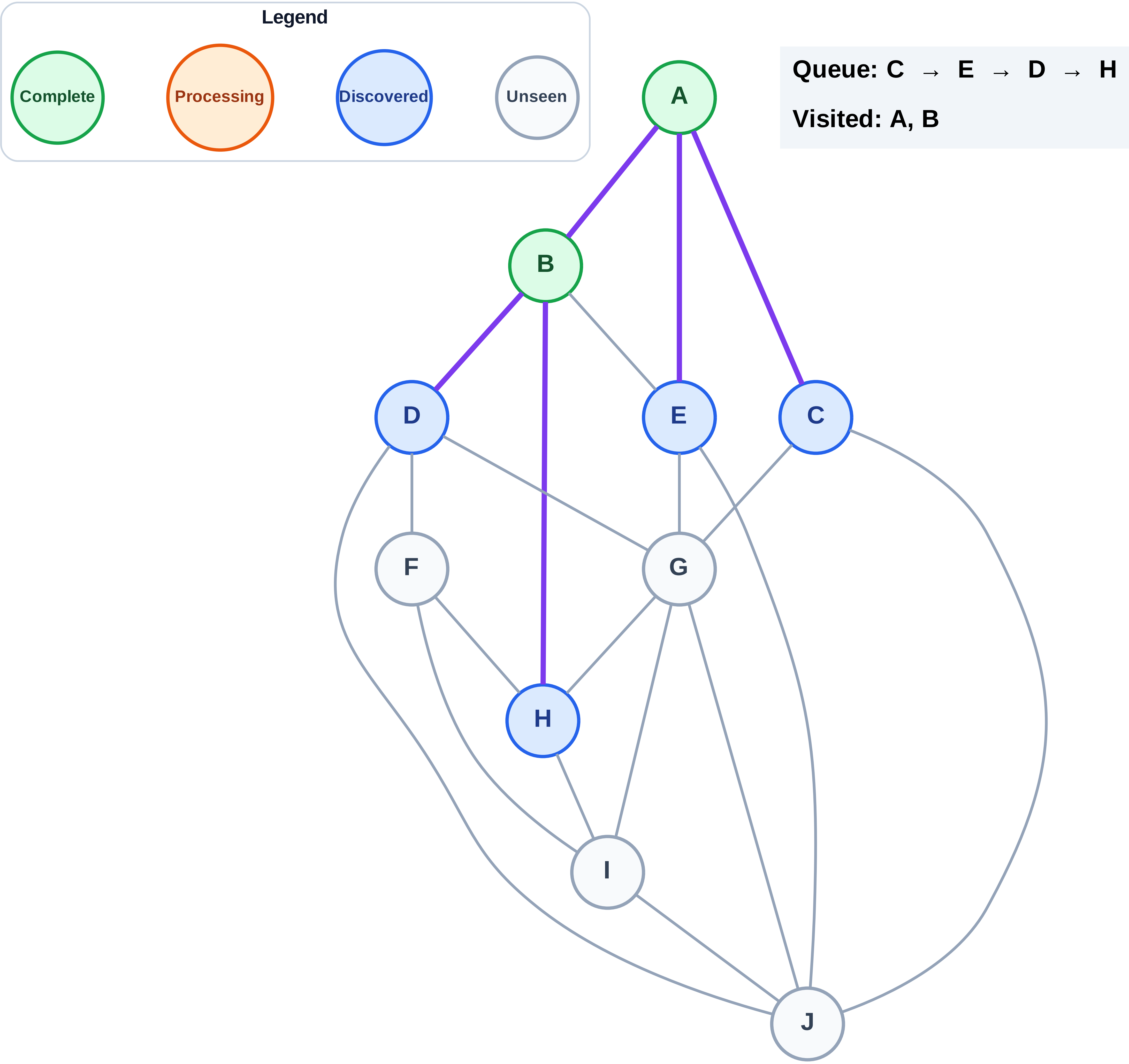
BFS Traversal

Step 4: Process node B



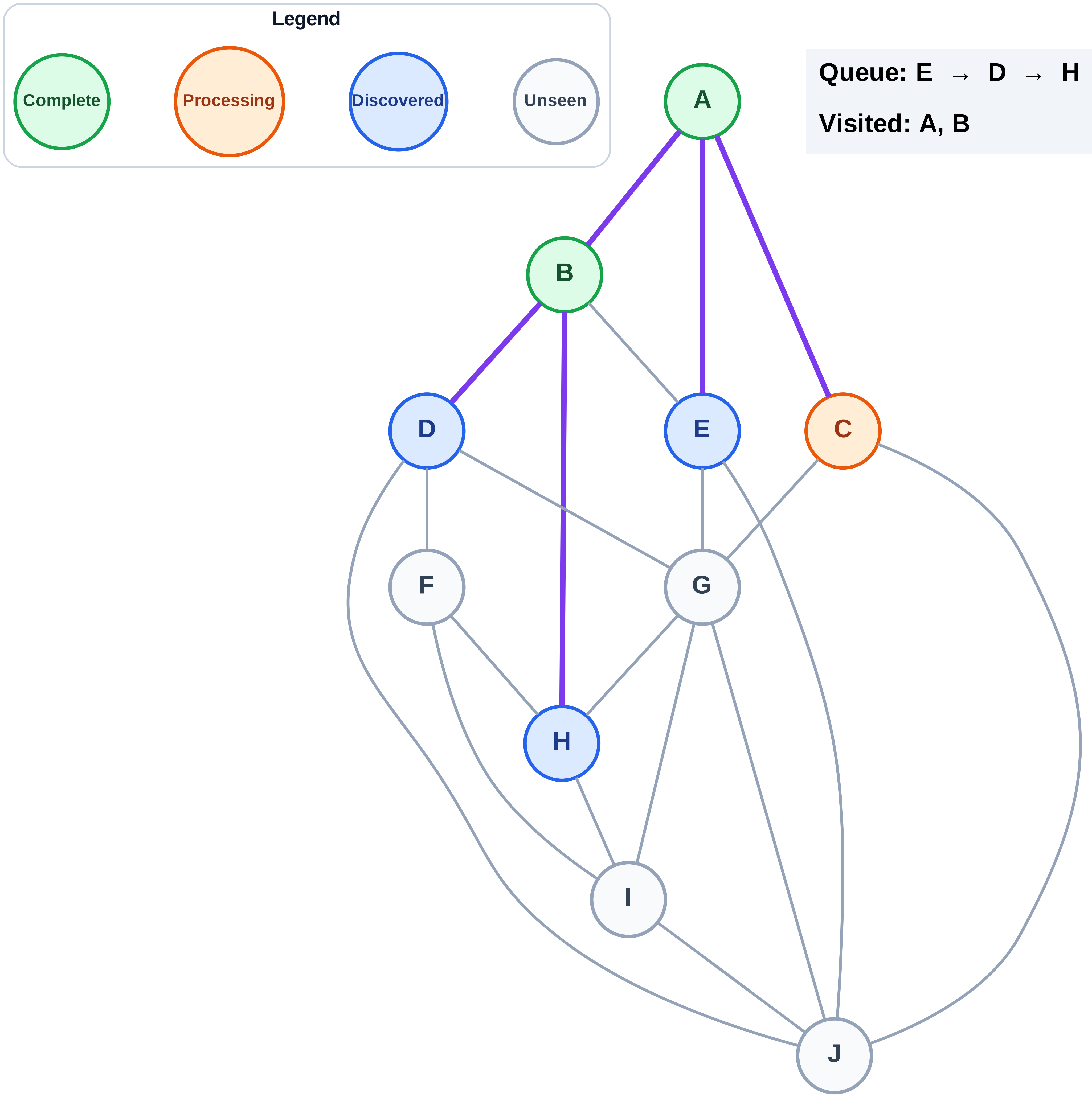
BFS Traversal

Step 5: Discover D, H from B



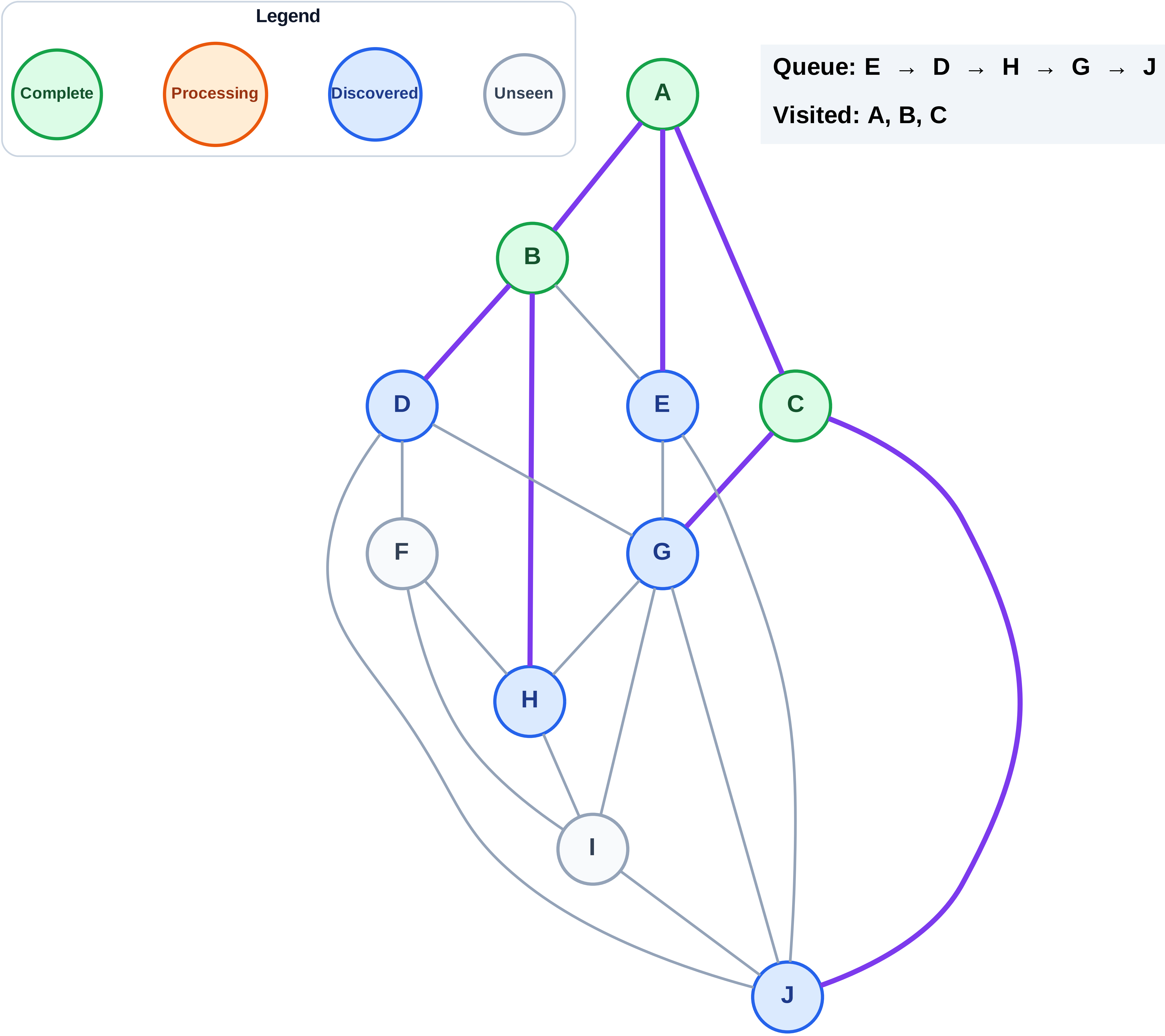
BFS Traversal

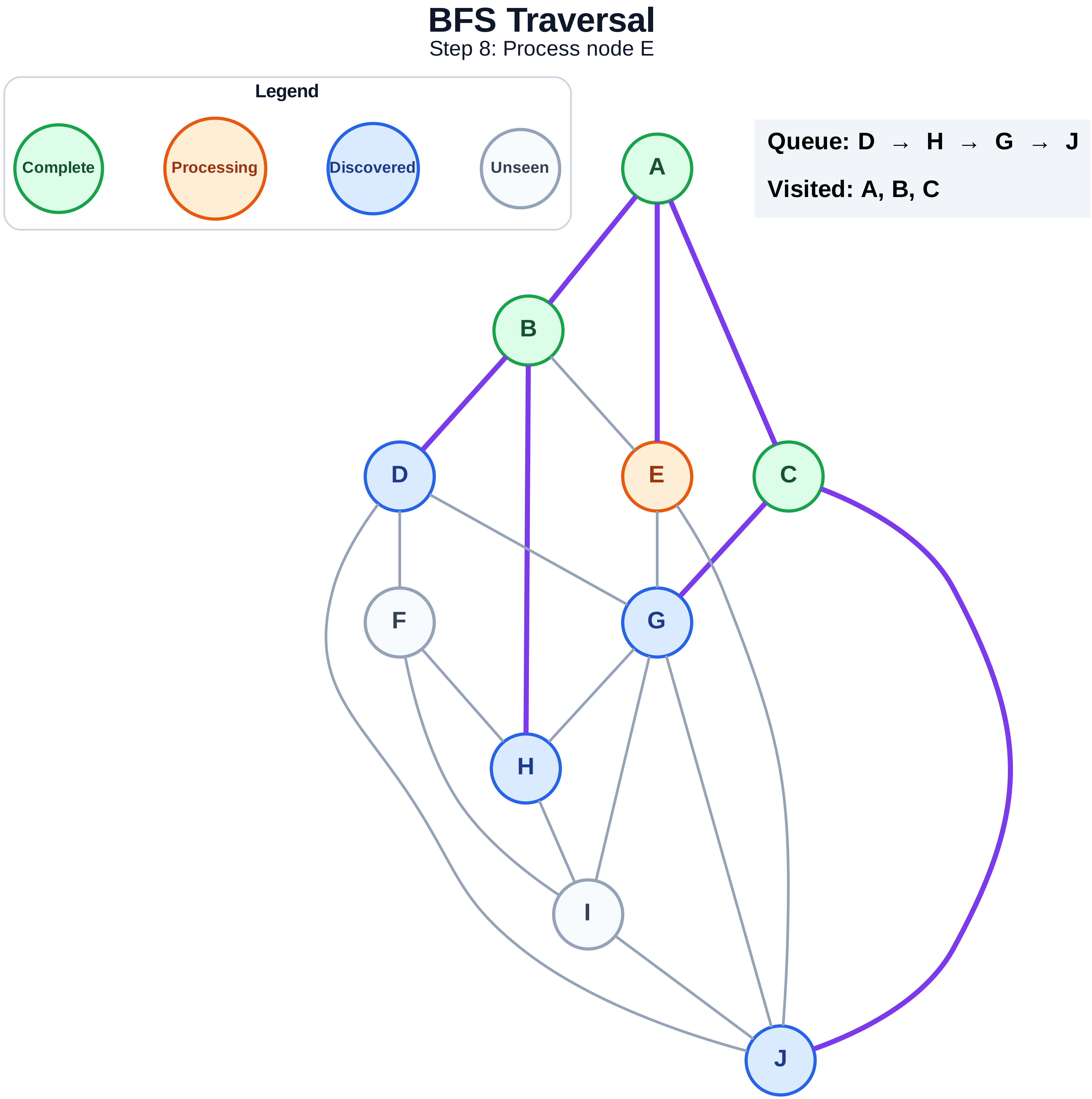
Step 6: Process node C



BFS Traversal

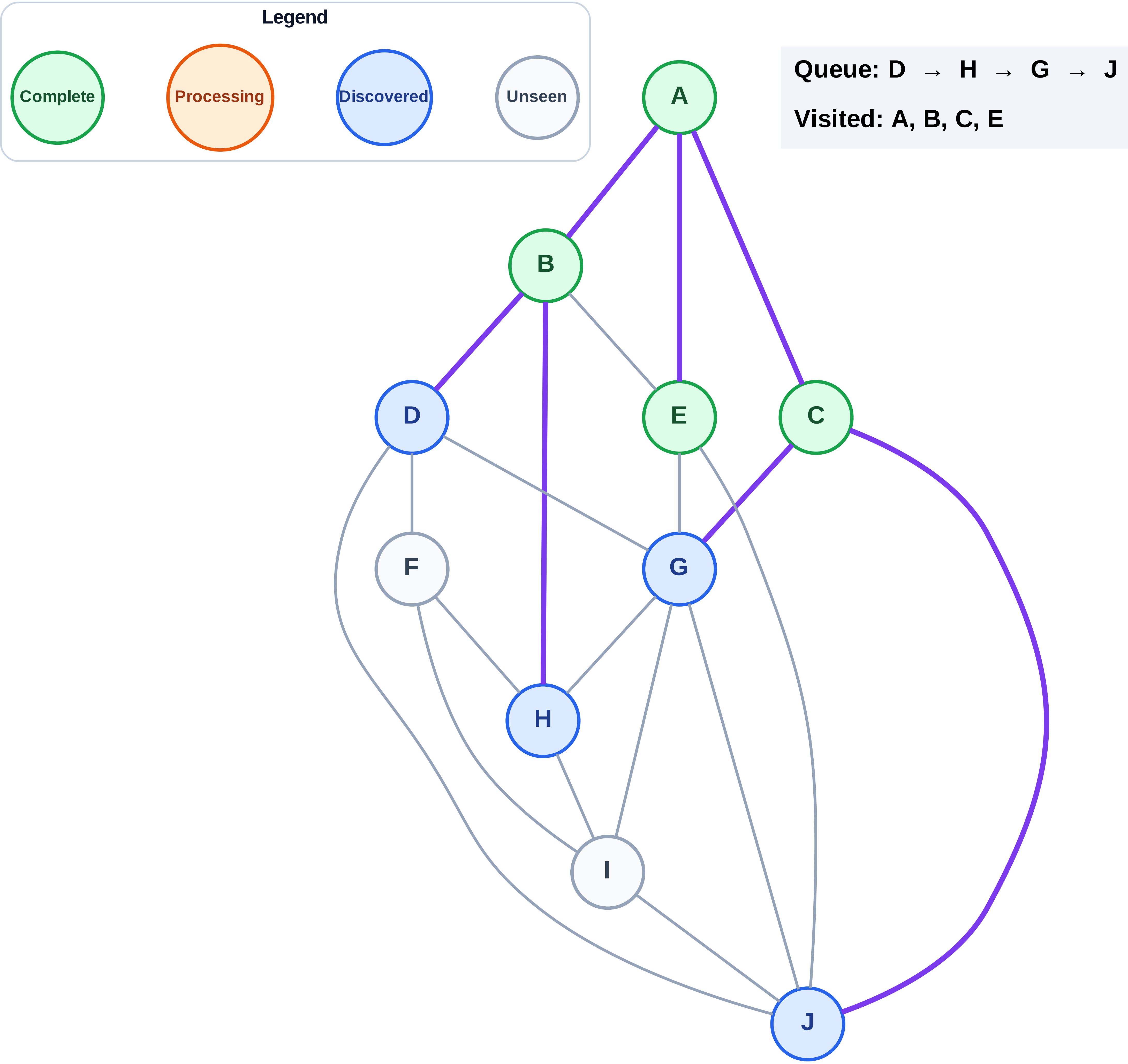
Step 7: Discover G, J from C




BFS Traversal
Step 8: Process node E
Legend
Complete
Processing
Discovered
Unseen
Queue: D → H → G → J
Visited: A, B, C
A
B
C
D
E
F
G
H
I
J

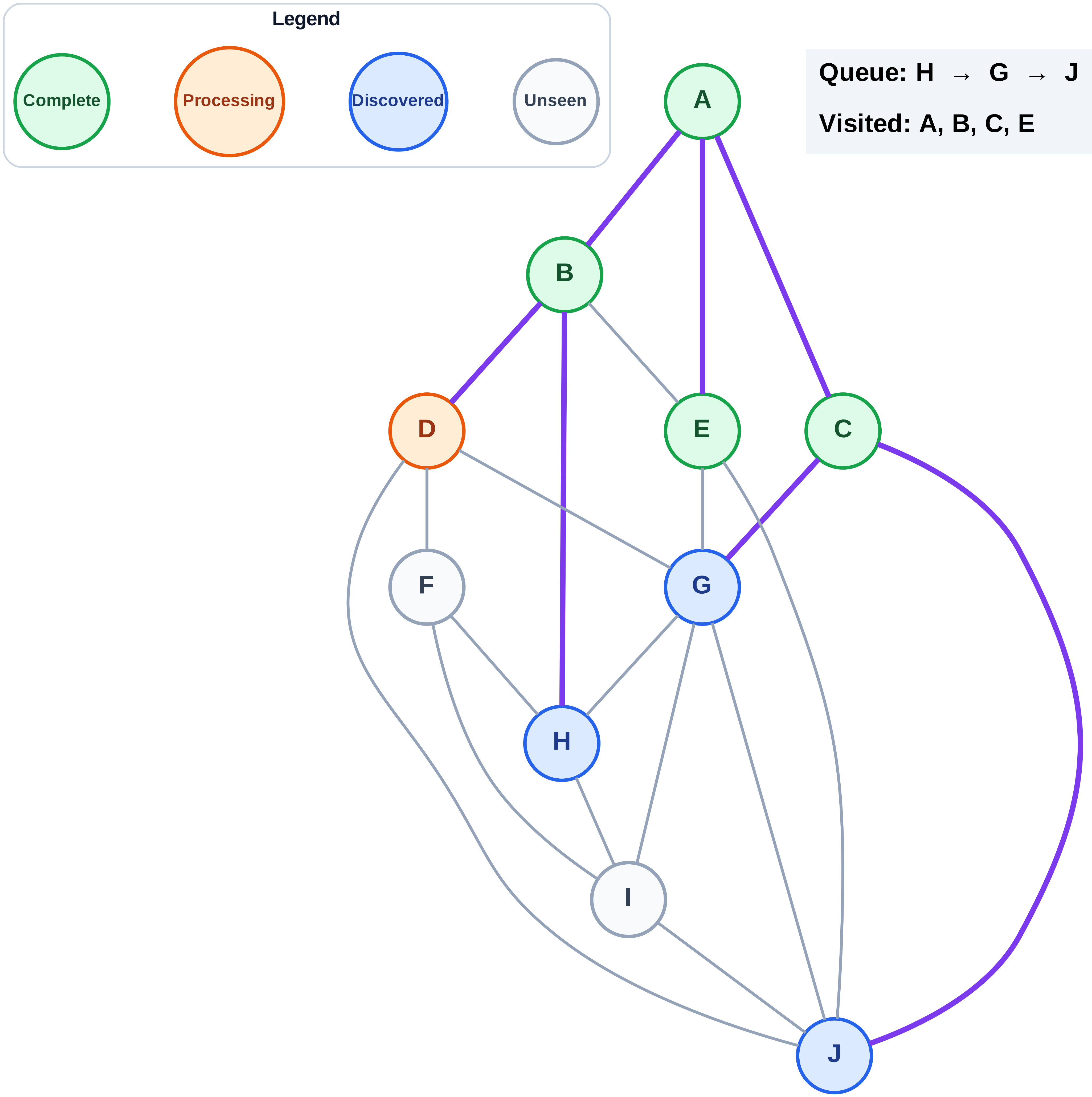
BFS Traversal

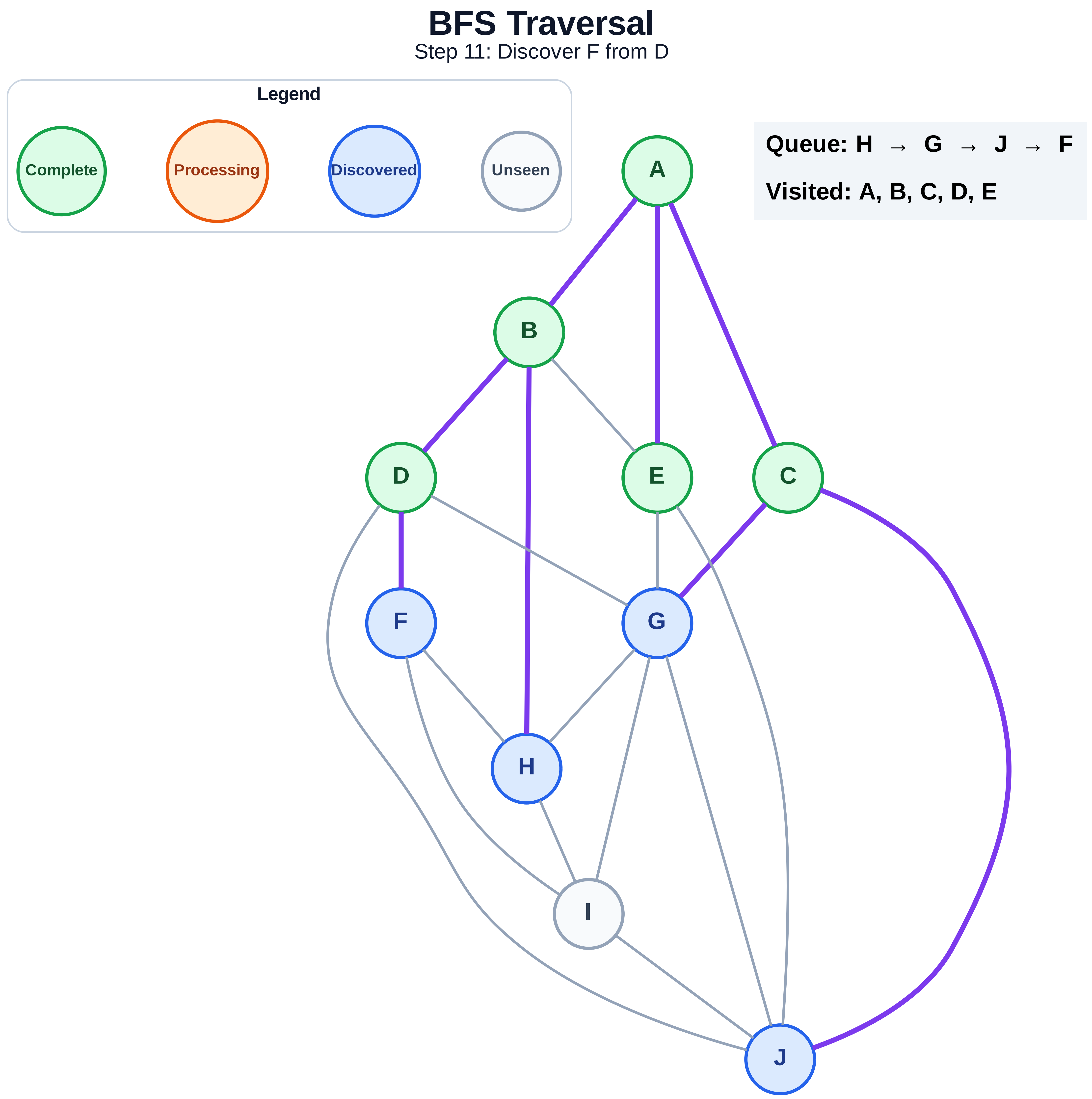
Step 9: No new neighbors from E



BFS Traversal

Step 10: Process node D

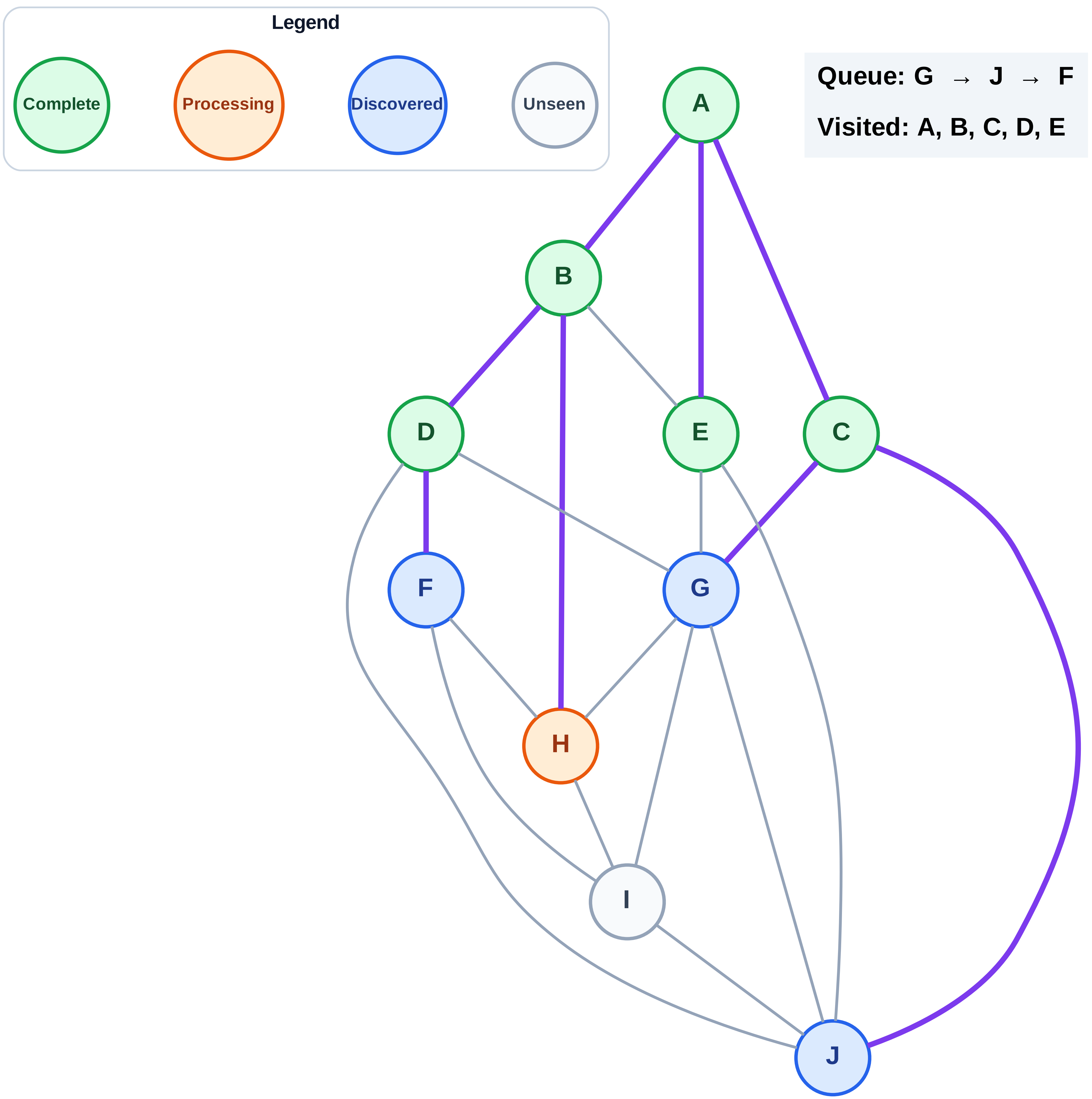




BFS Traversal
Step 11: Discover F from D
Legend
Complete
Processing
Discovered
Unseen
Queue: H → G → J → F
Visited: A, B, C, D, E
A
B
C
D
E
F
G
H
I
J

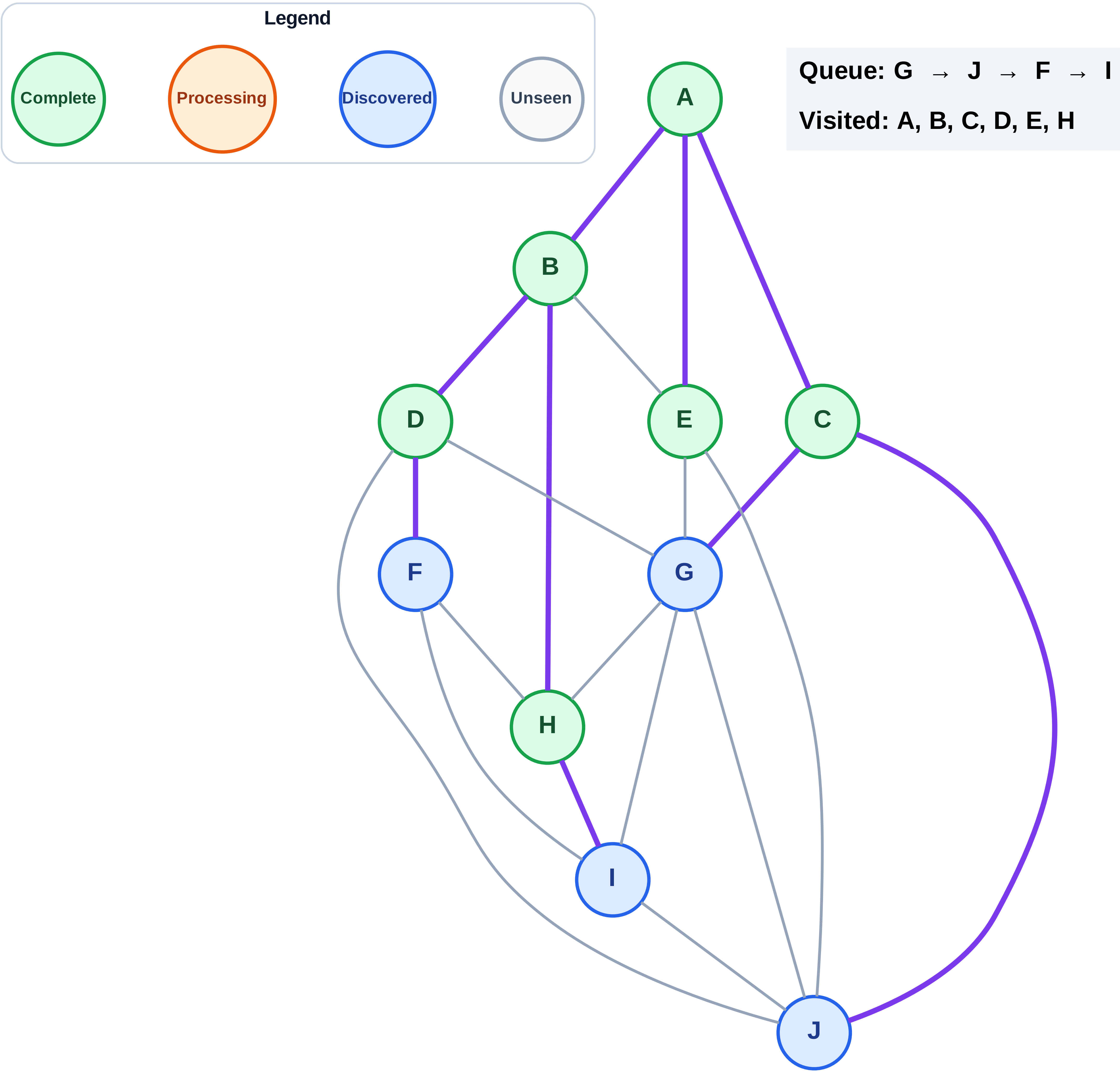
BFS Traversal

Step 12: Process node H



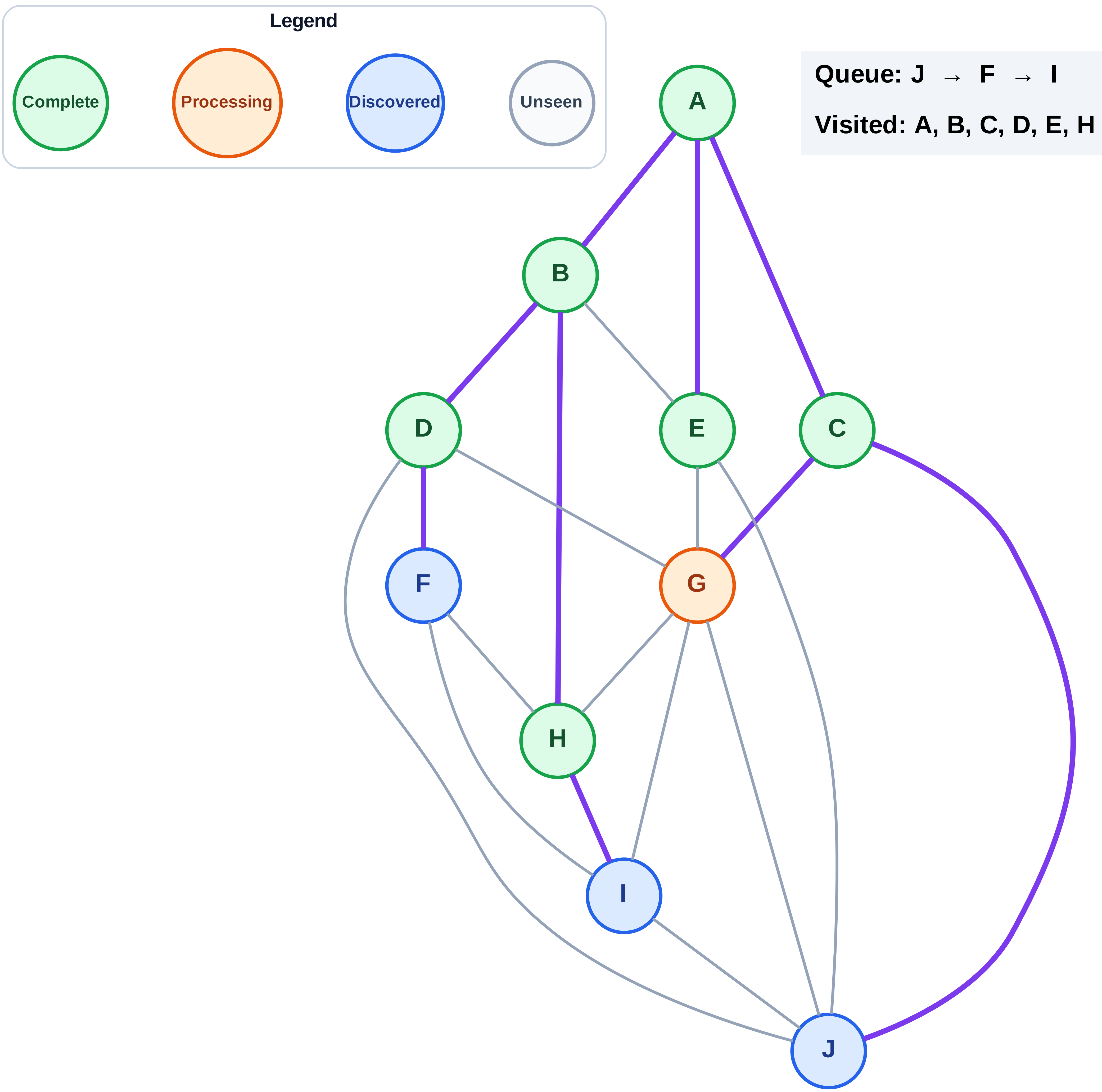
BFS Traversal

Step 13: Discover I from H



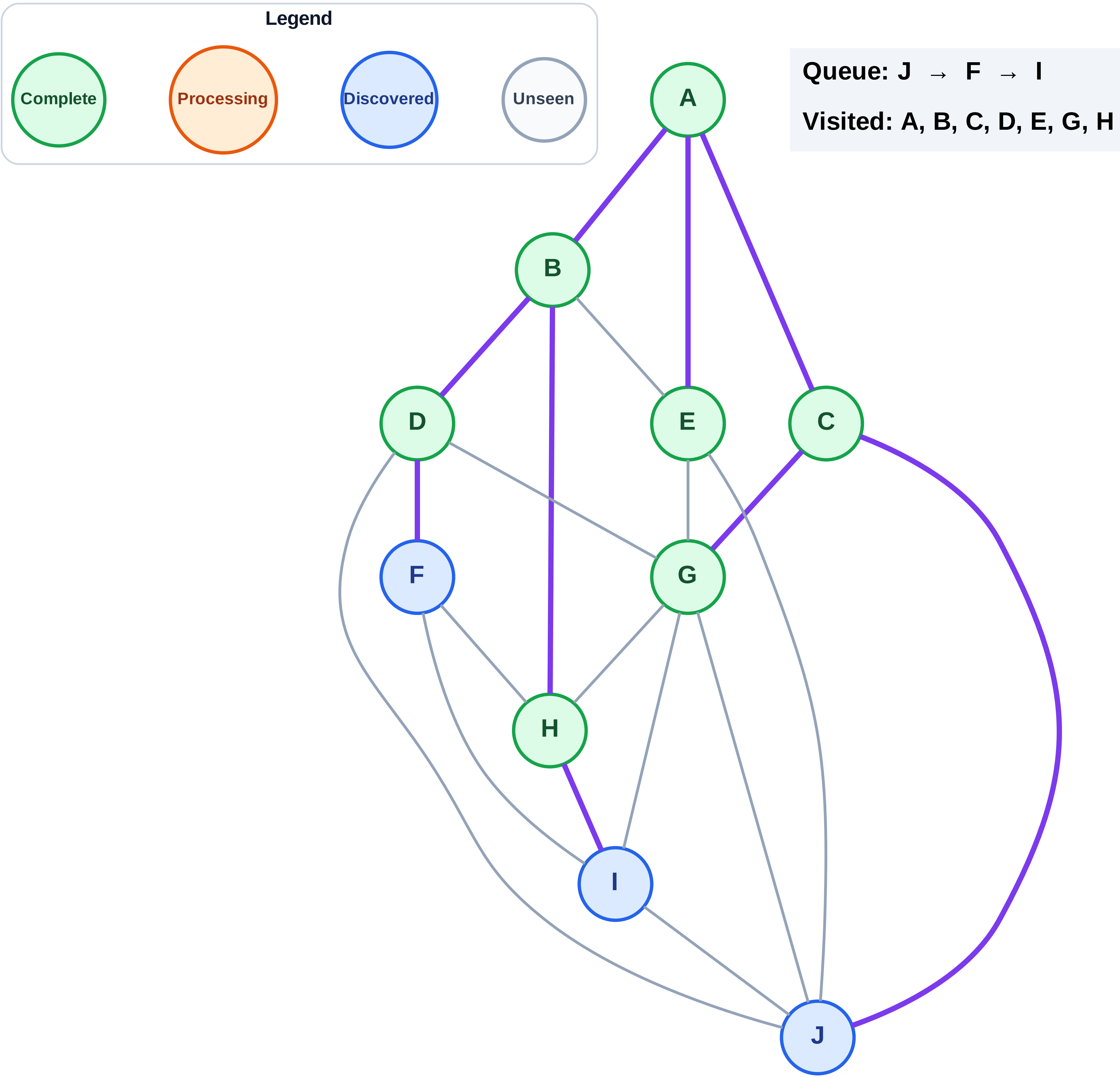
BFS Traversal

Step 14: Process node G



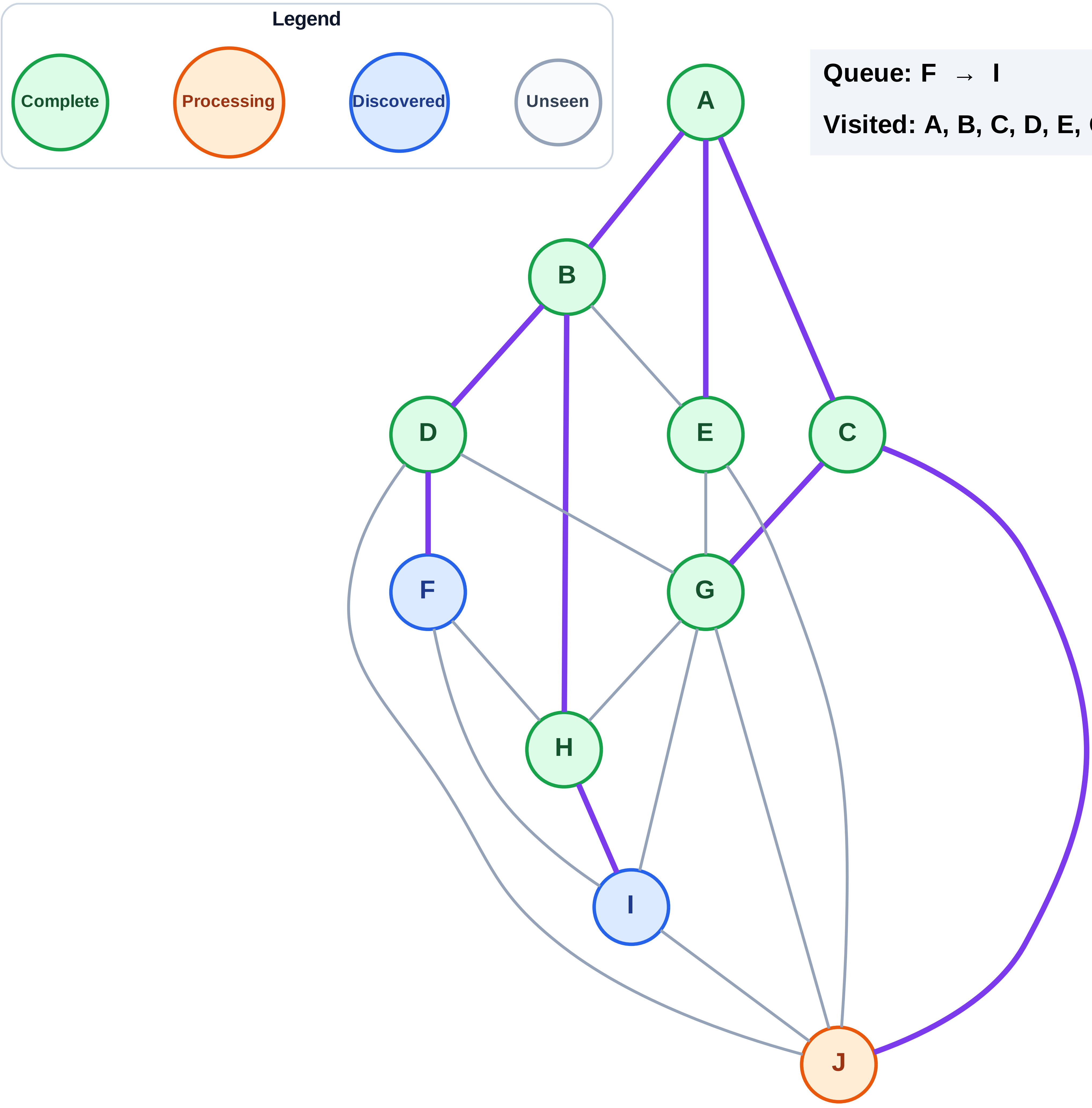
BFS Traversal

Step 15: No new neighbors from G



BFS Traversal

Step 16: Process node J

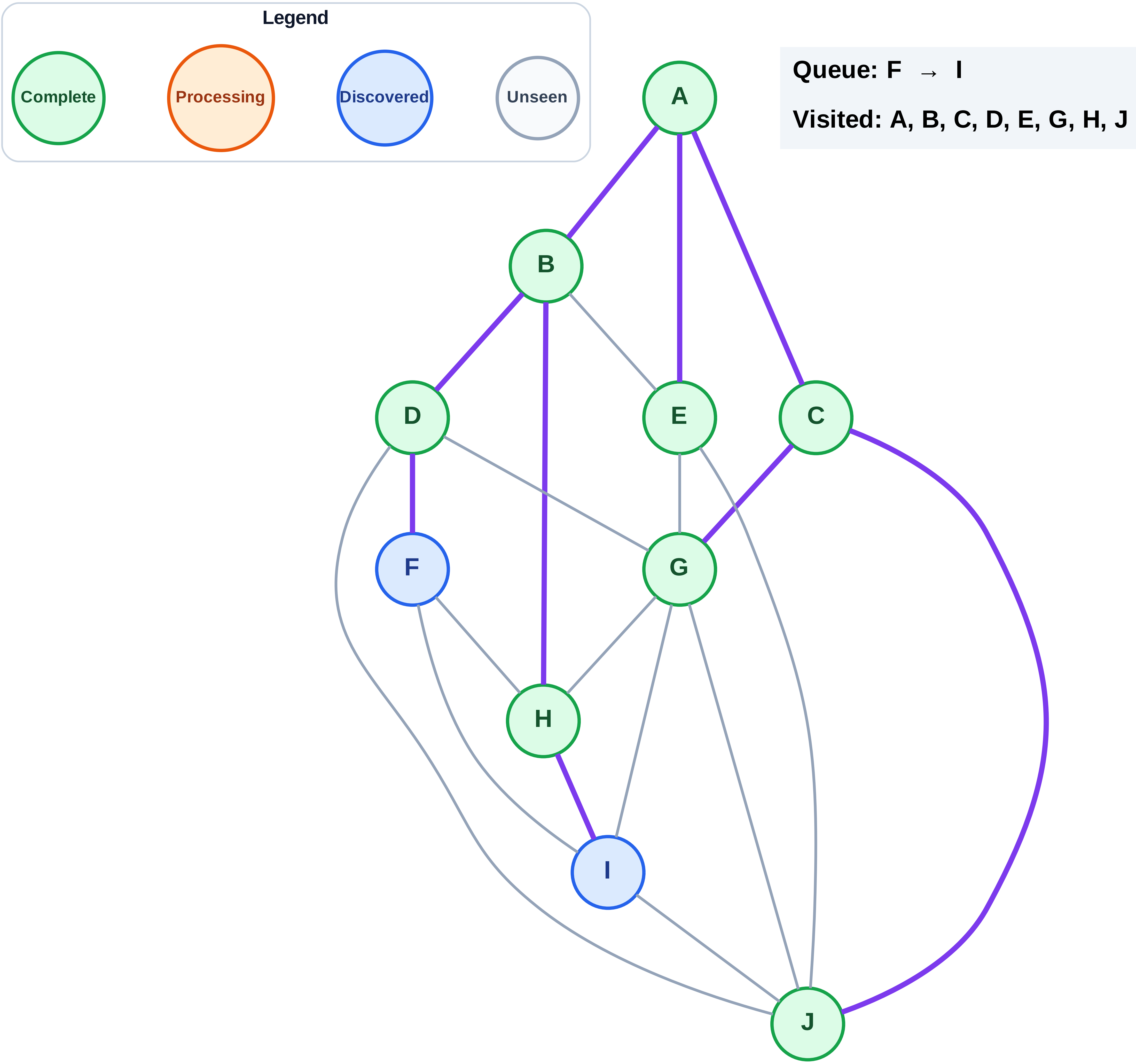


Queue: F → I

Visited: A, B, C, D, E, G, H

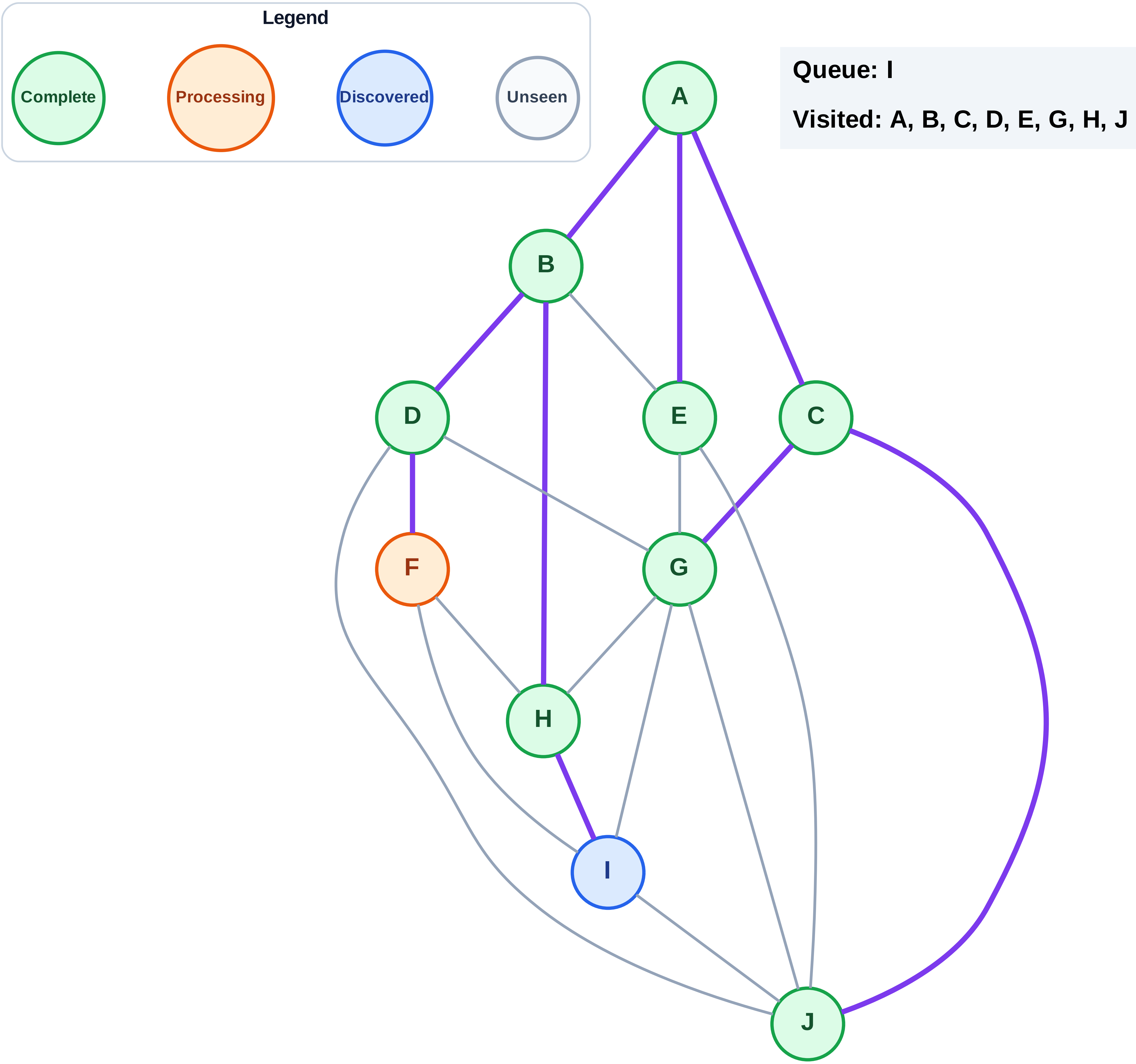
BFS Traversal

Step 17: No new neighbors from J



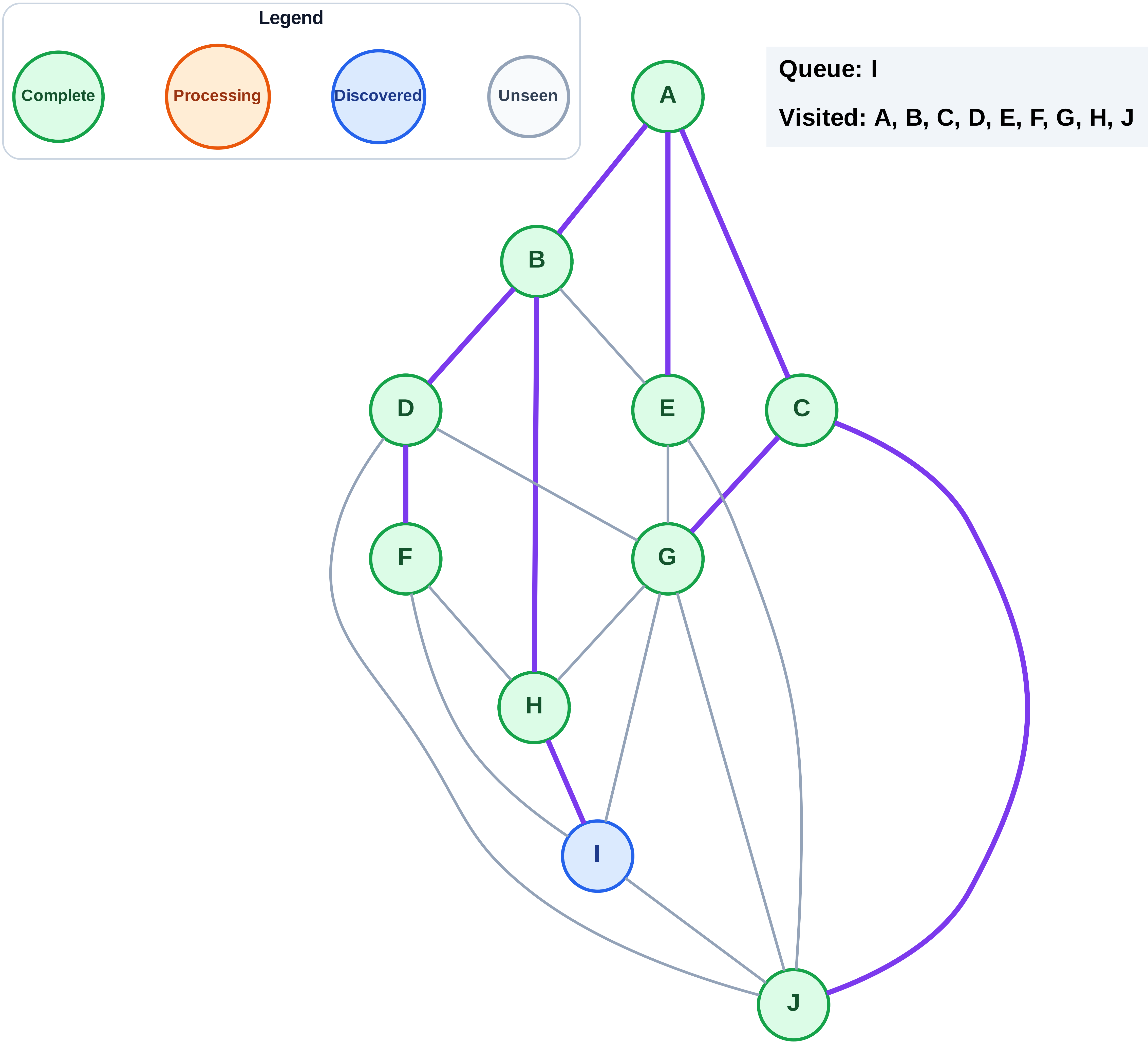
BFS Traversal

Step 18: Process node F



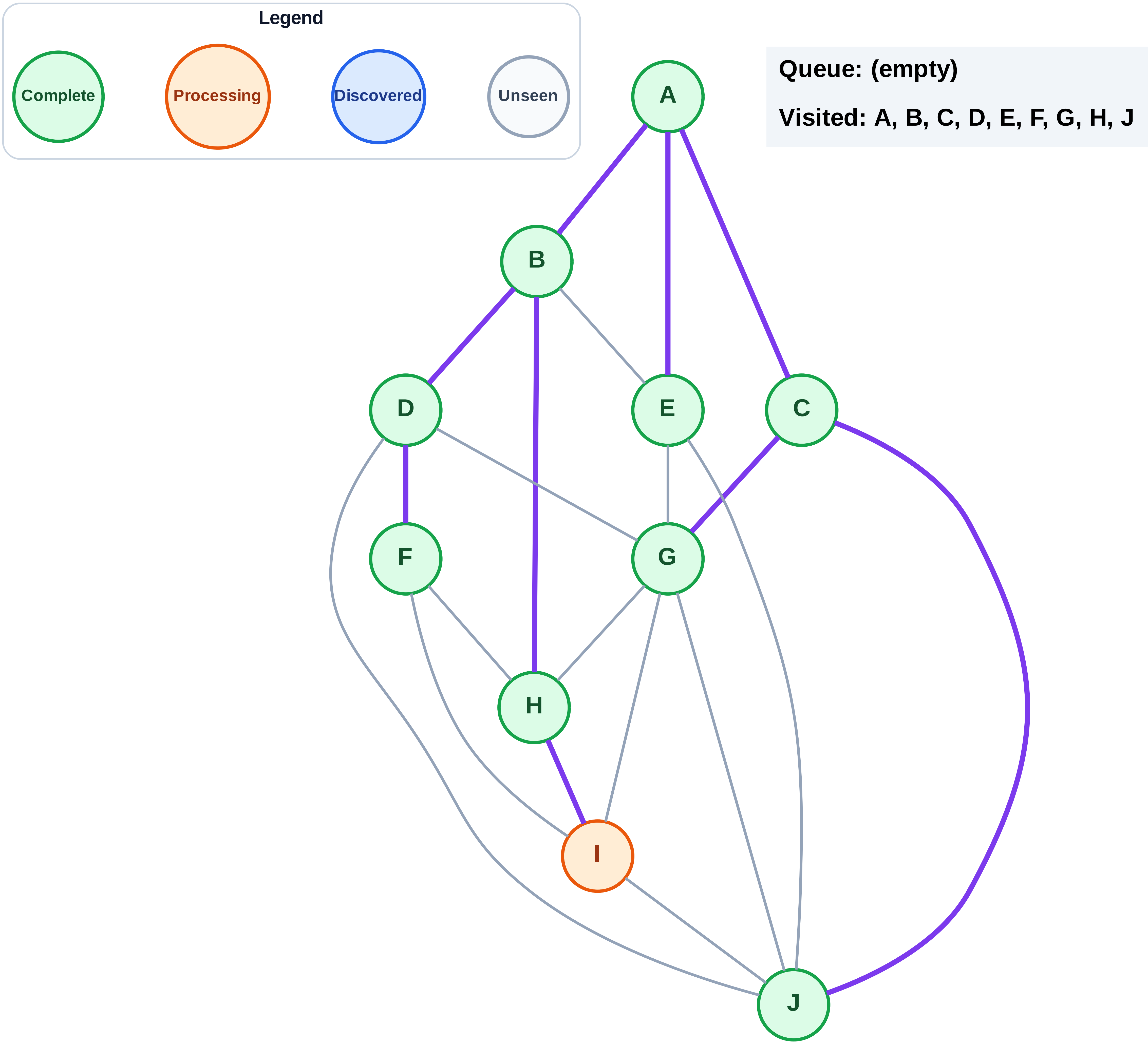
BFS Traversal

Step 19: No new neighbors from F



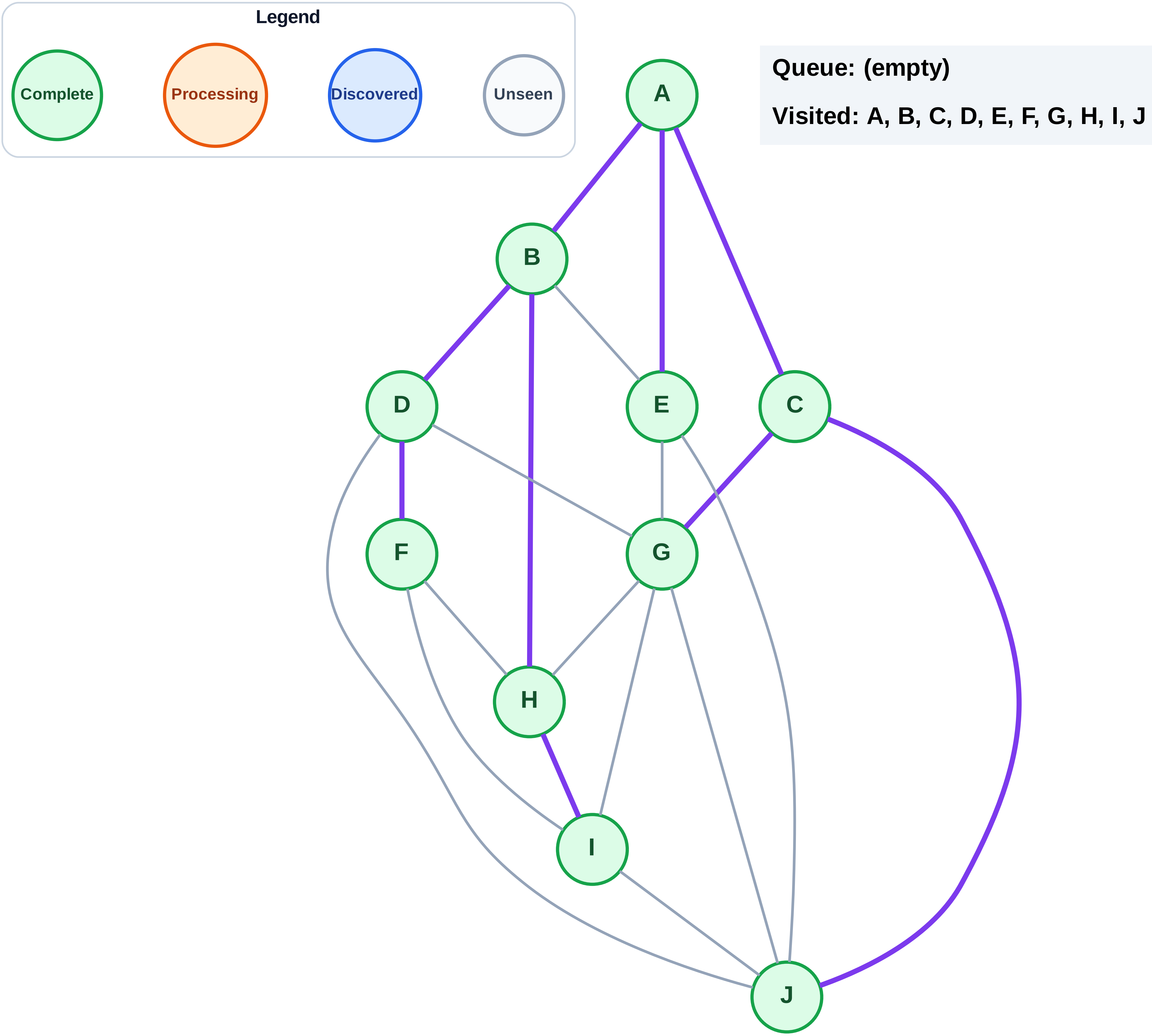
BFS Traversal

Step 20: Process node I



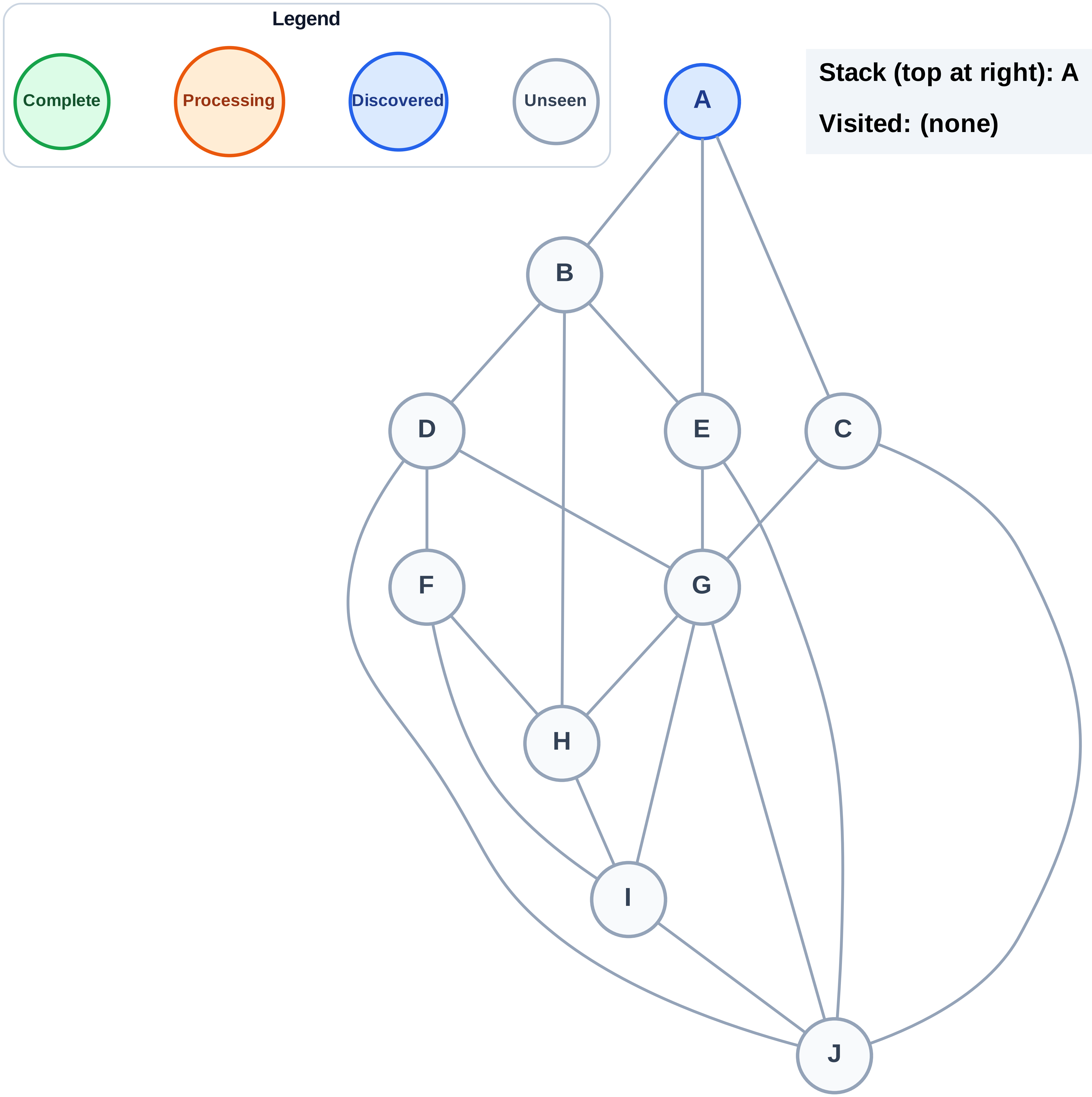
BFS Traversal

Step 21: No new neighbors from I



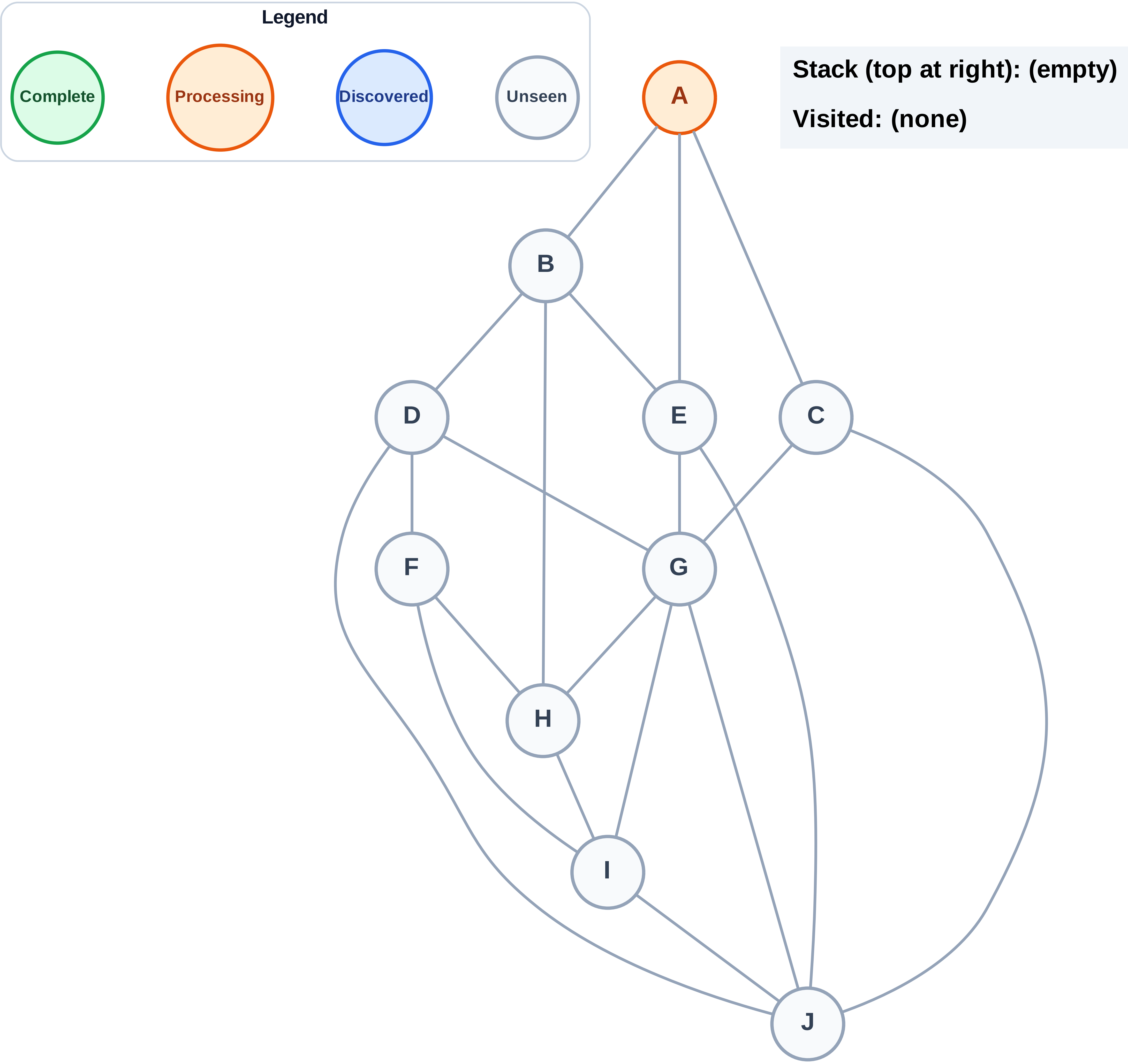
DFS Traversal

Step 1: Start at A



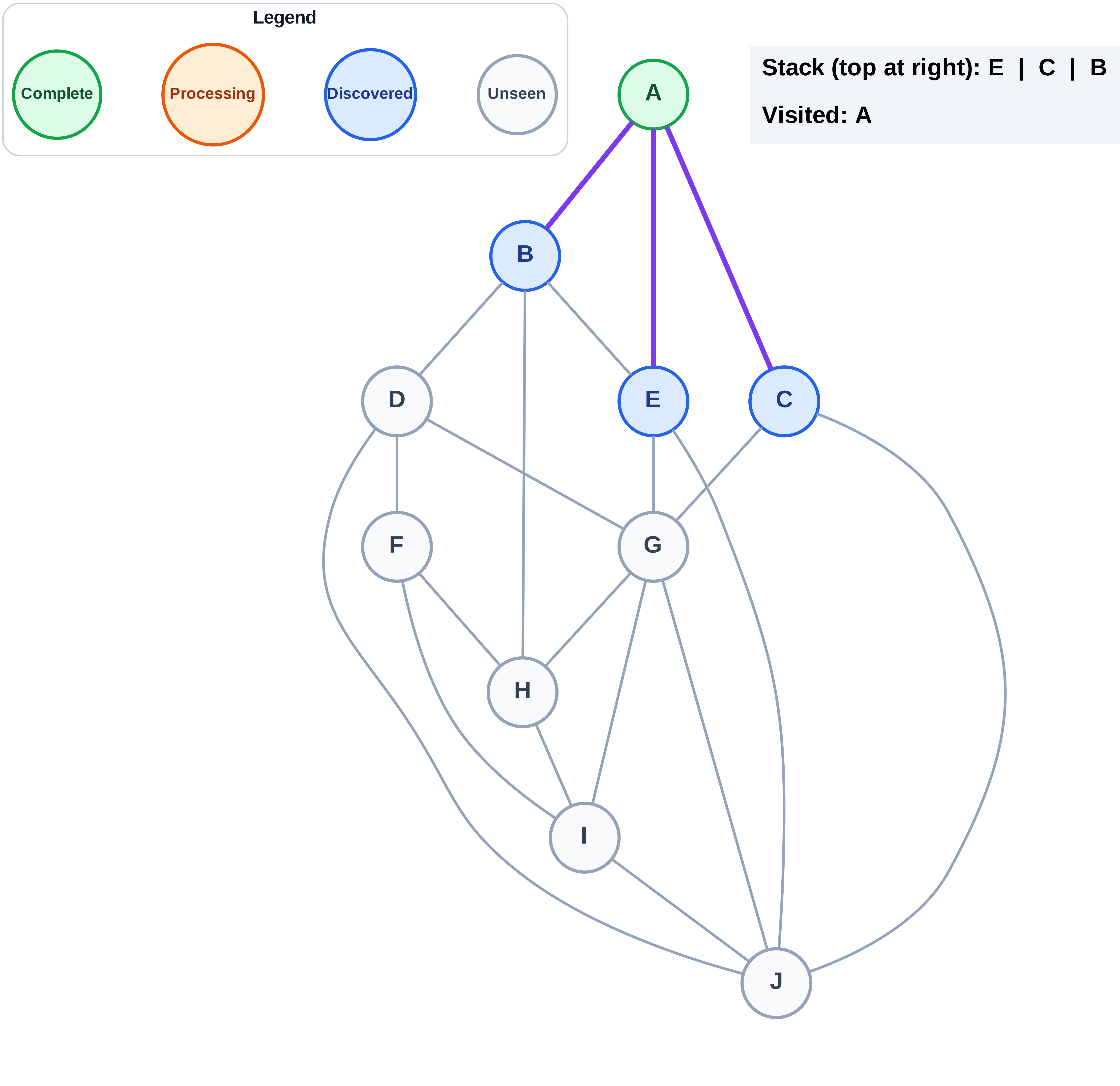
DFS Traversal

Step 2: Process node A



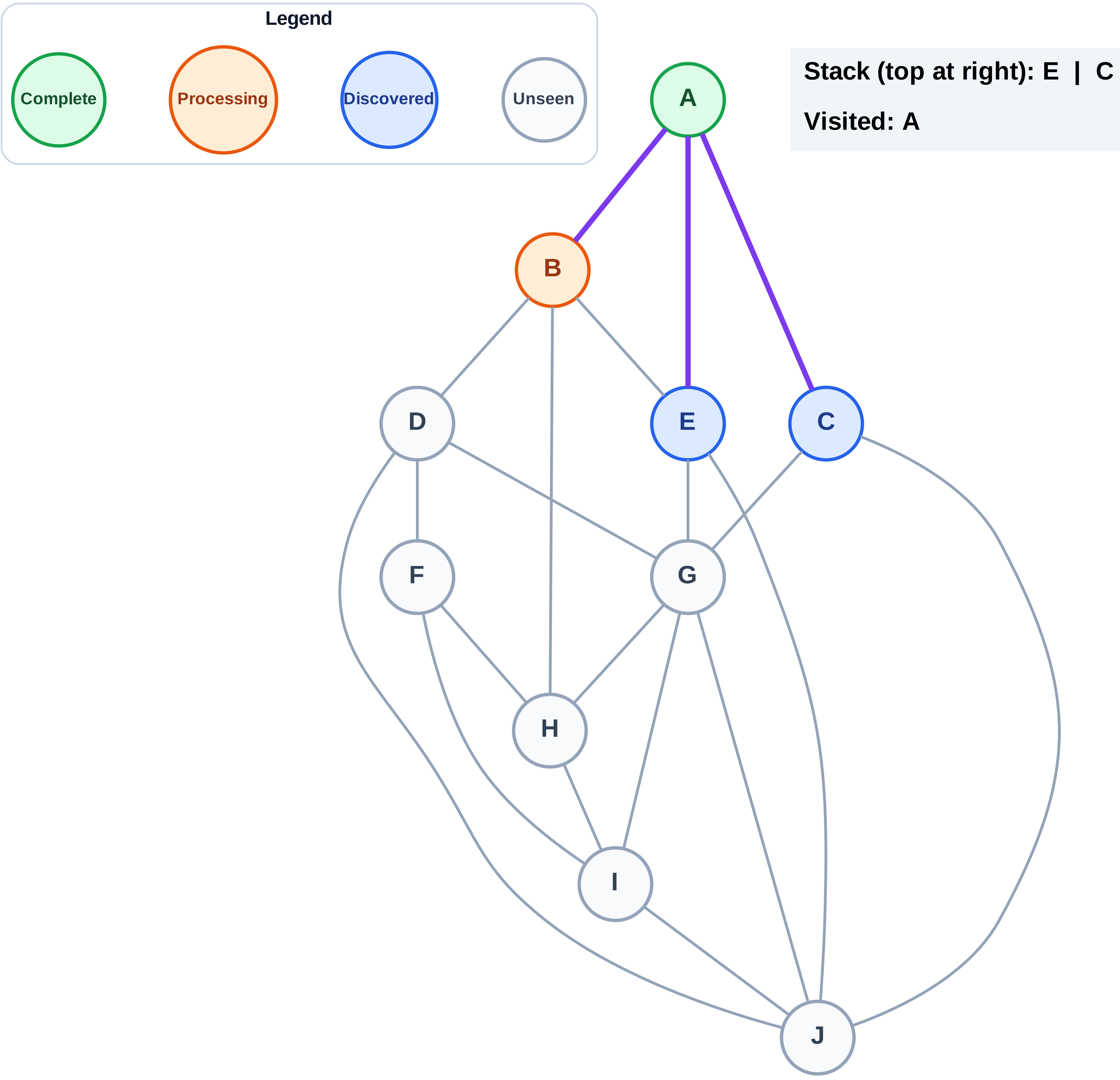
DFS Traversal

Step 3: Discover E, C, B from A



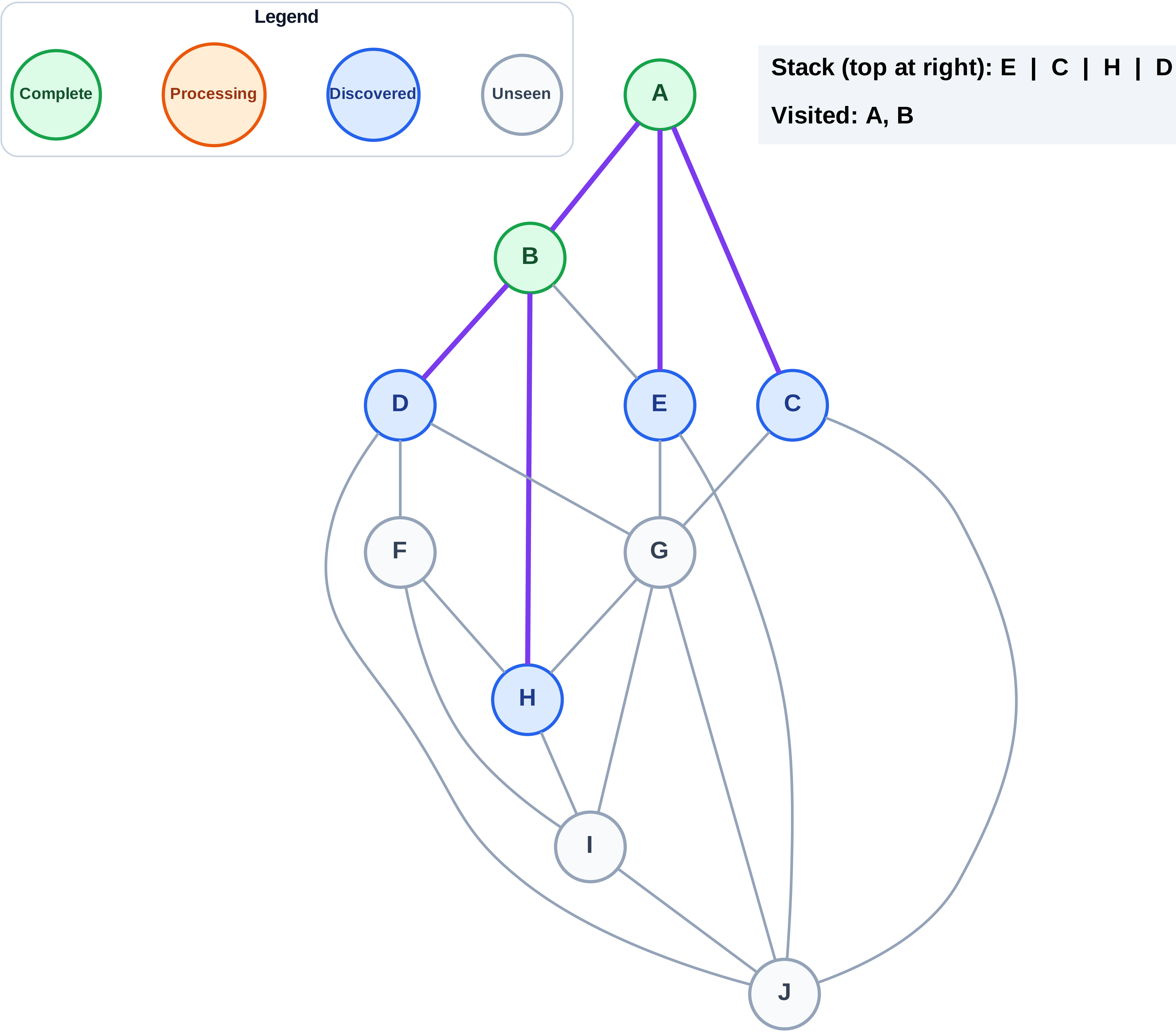
DFS Traversal

Step 4: Process node B



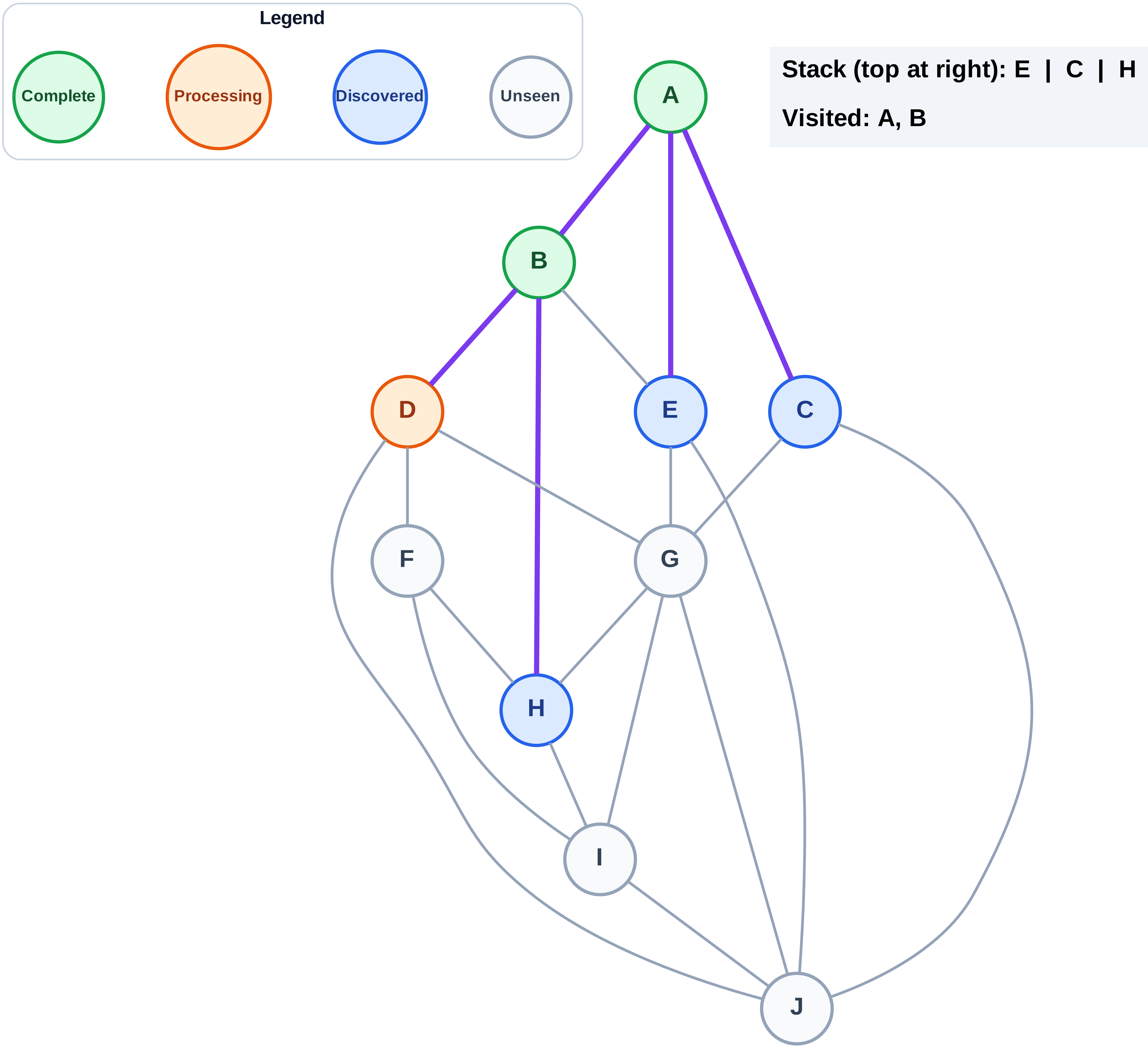
DFS Traversal

Step 5: Discover H, D from B



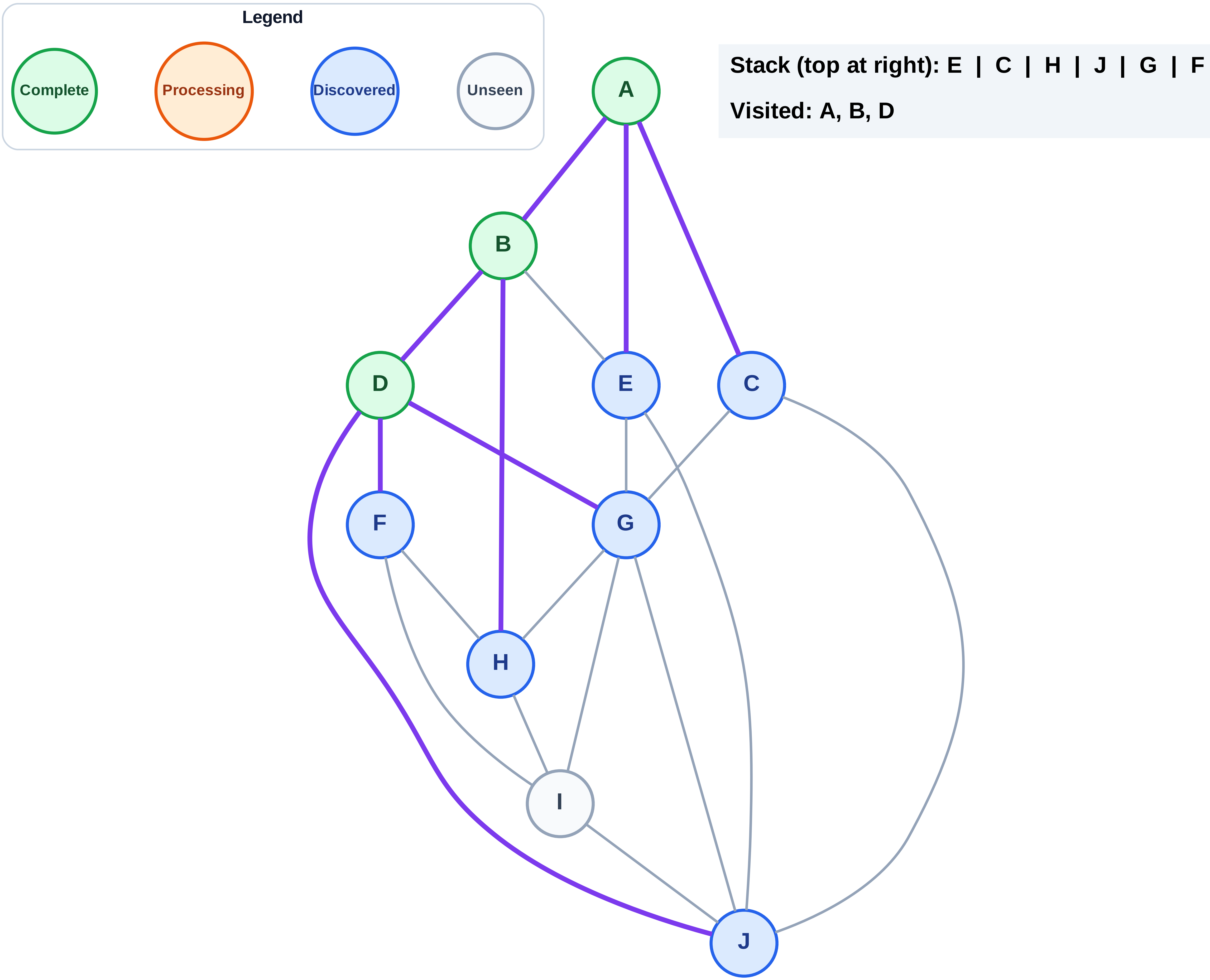
DFS Traversal

Step 6: Process node D



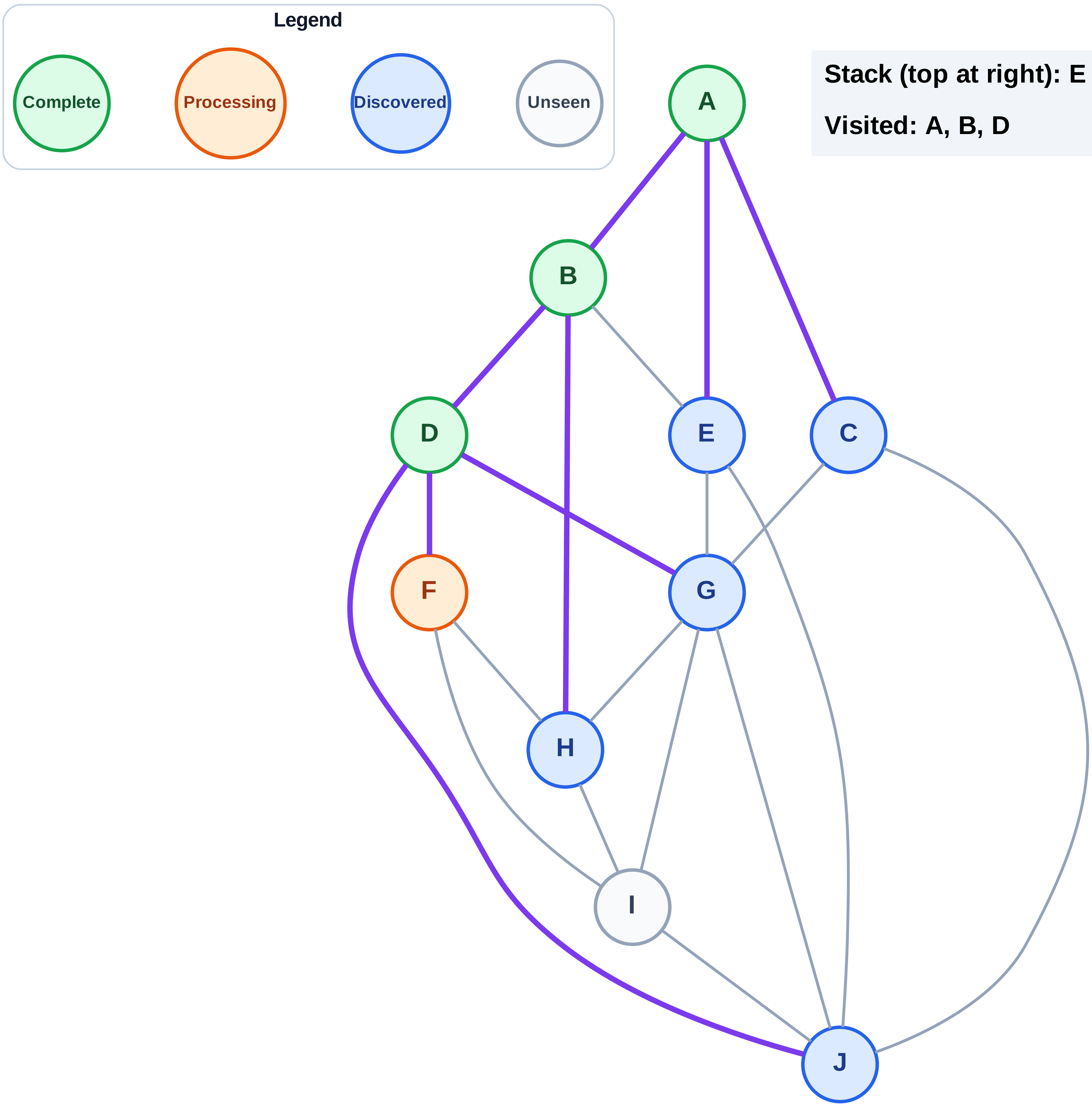
DFS Traversal

Step 7: Discover J, G, F from D



DFS Traversal

Step 8: Process node F

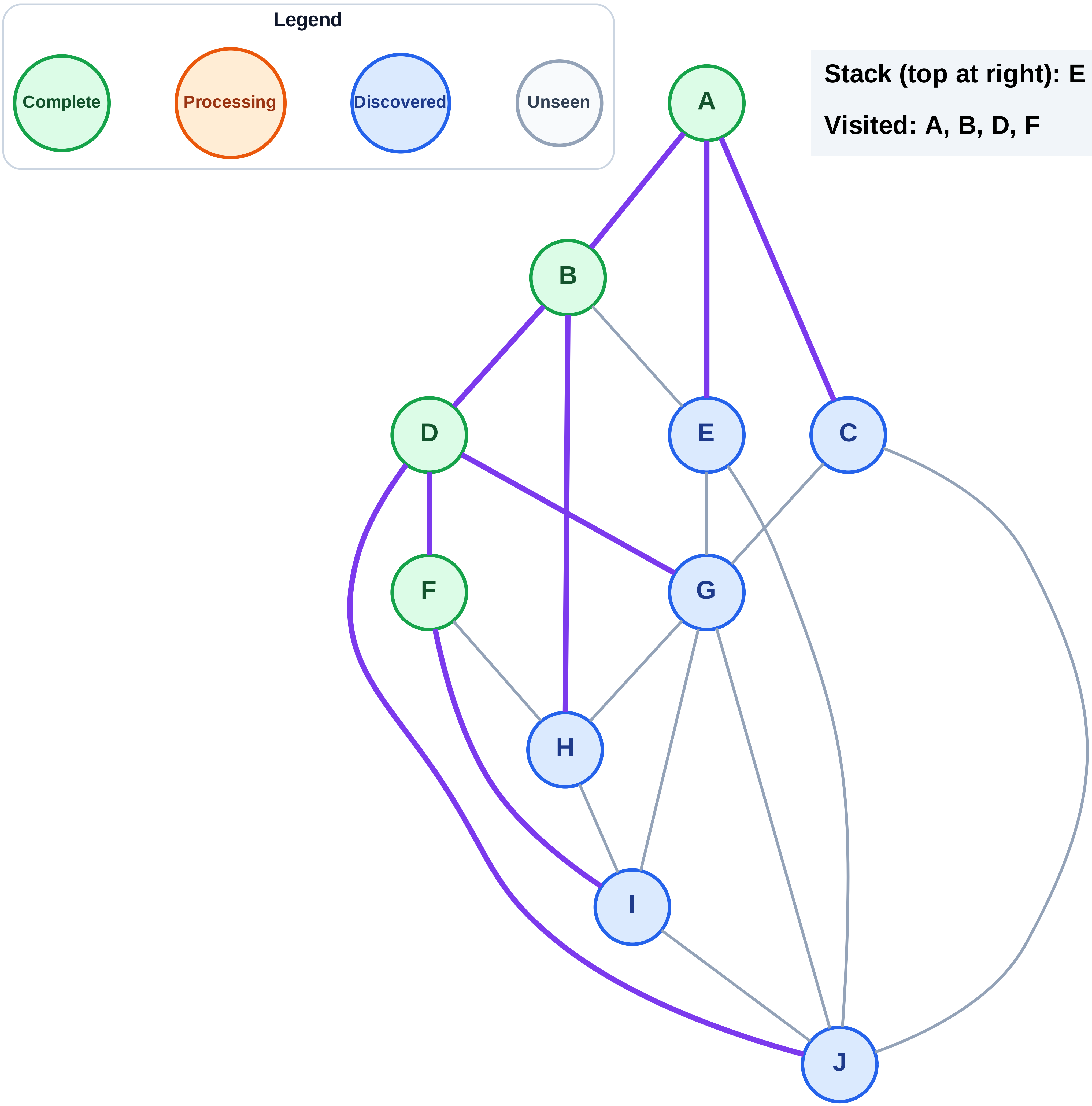


Stack (top at right): E | C | H | J | G

Visited: A, B, D

DFS Traversal

Step 9: Discover I from F

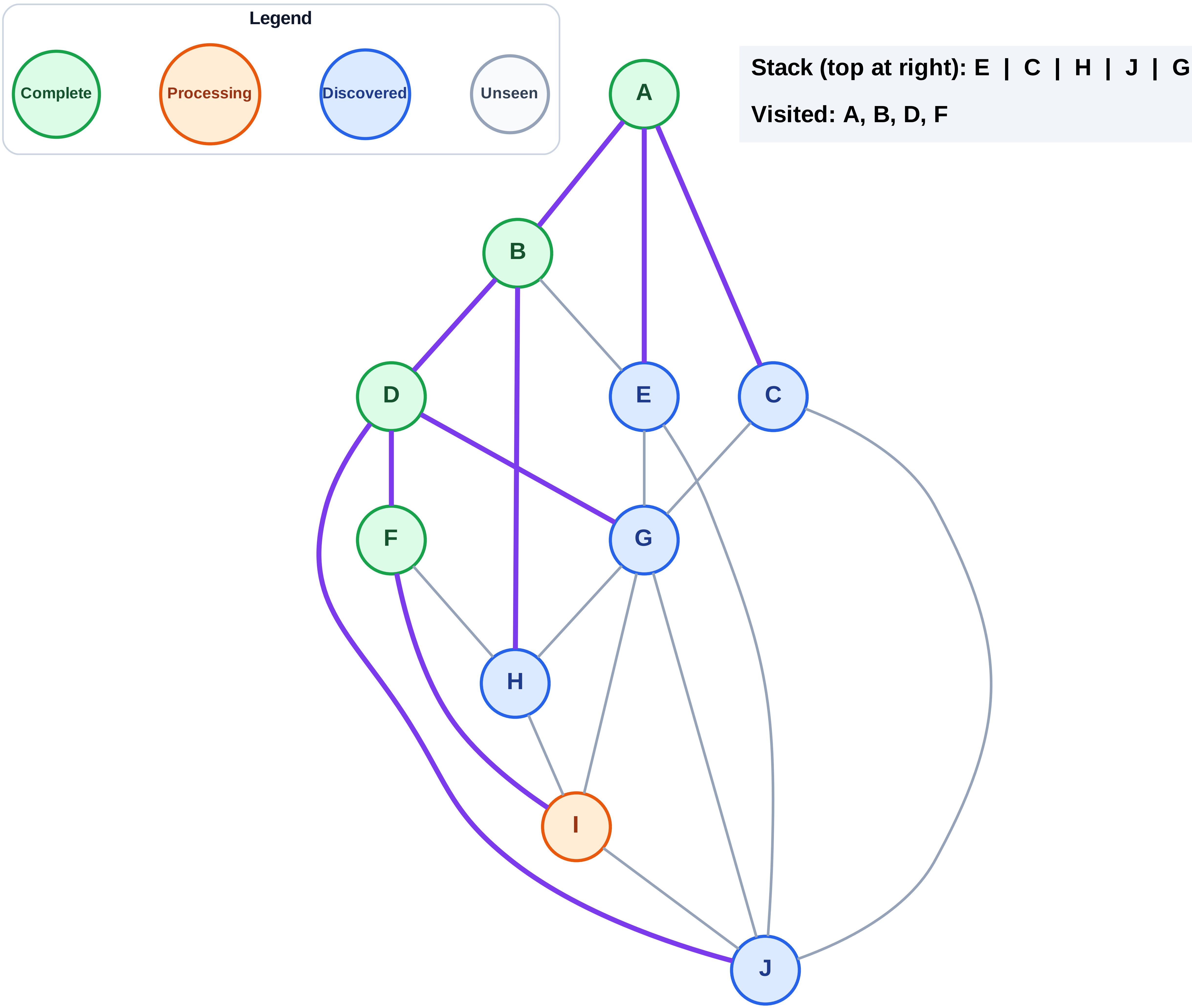


Stack (top at right): E | C | H | J | G | I

Visited: A, B, D, F

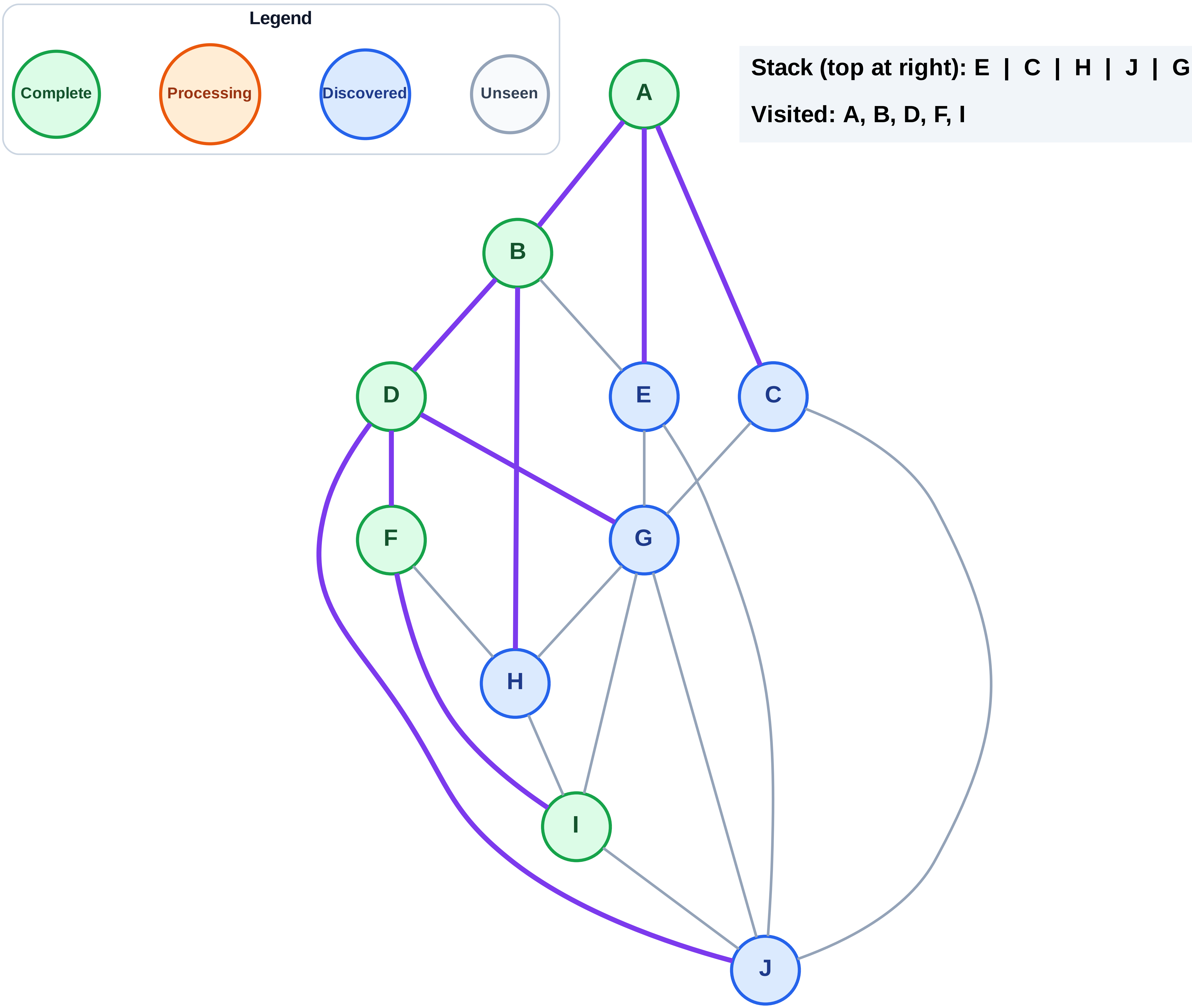
DFS Traversal

Step 10: Process node I



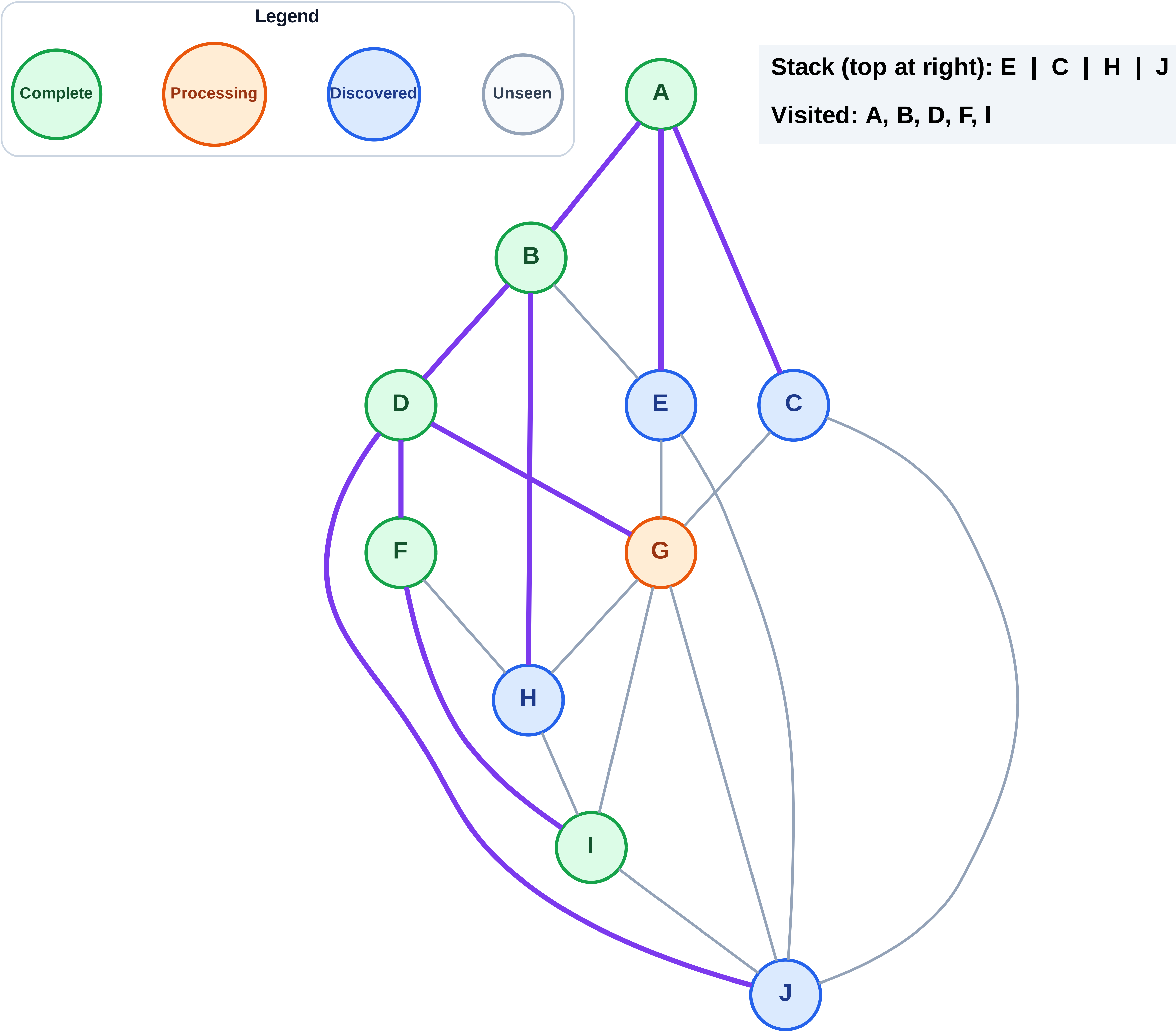
DFS Traversal

Step 11: No new neighbors from I



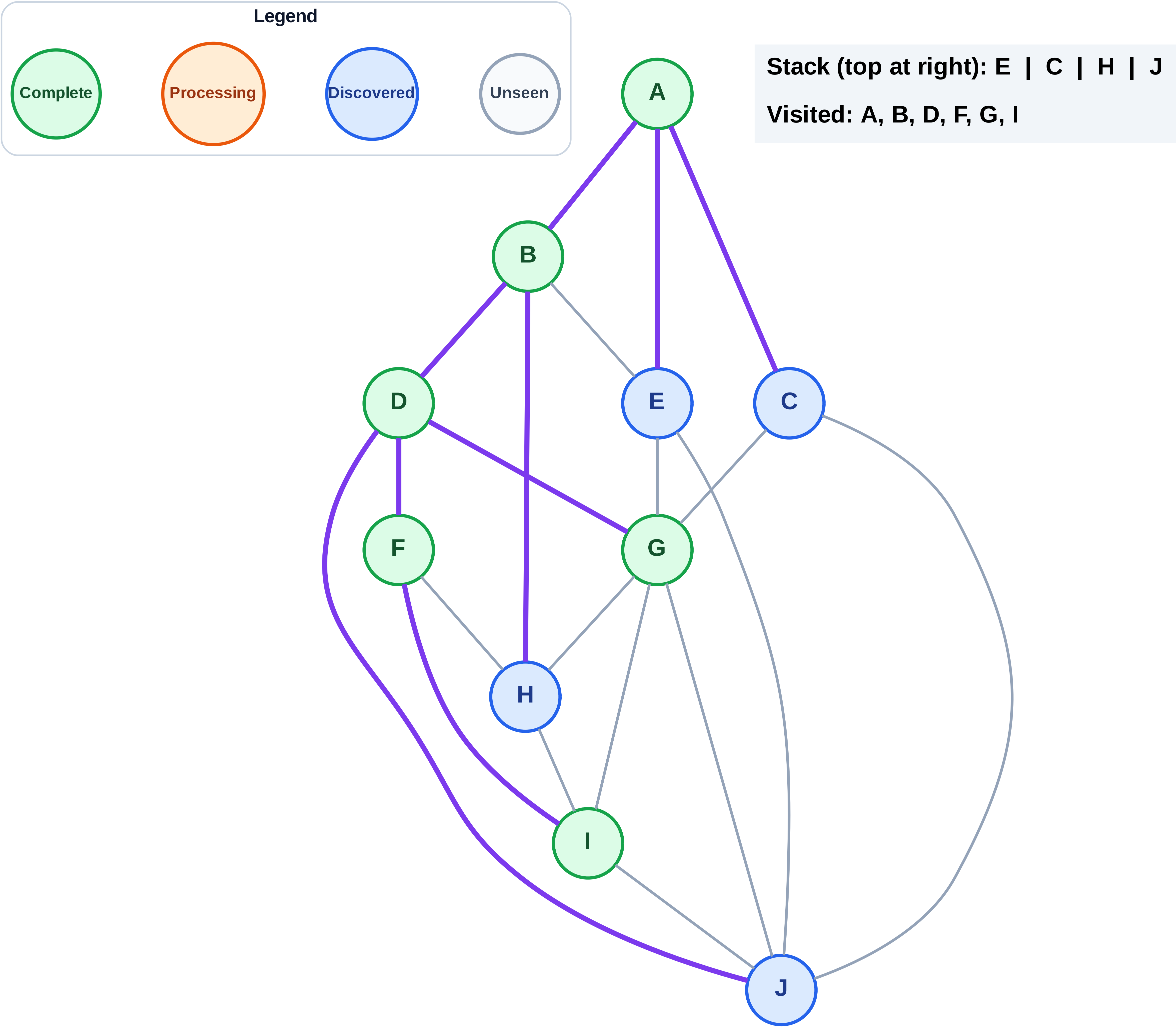
DFS Traversal

Step 12: Process node G



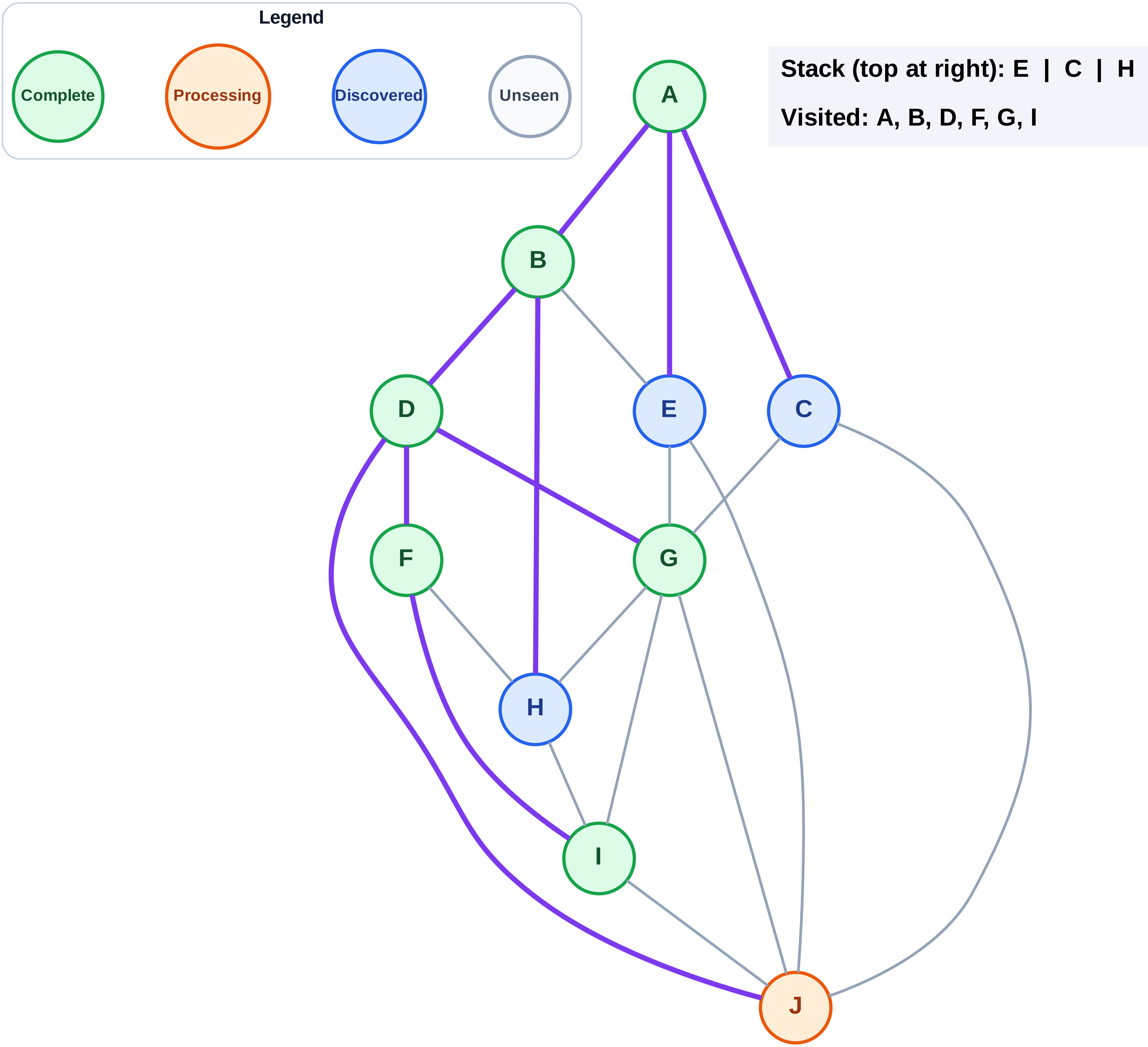
DFS Traversal

Step 13: No new neighbors from G



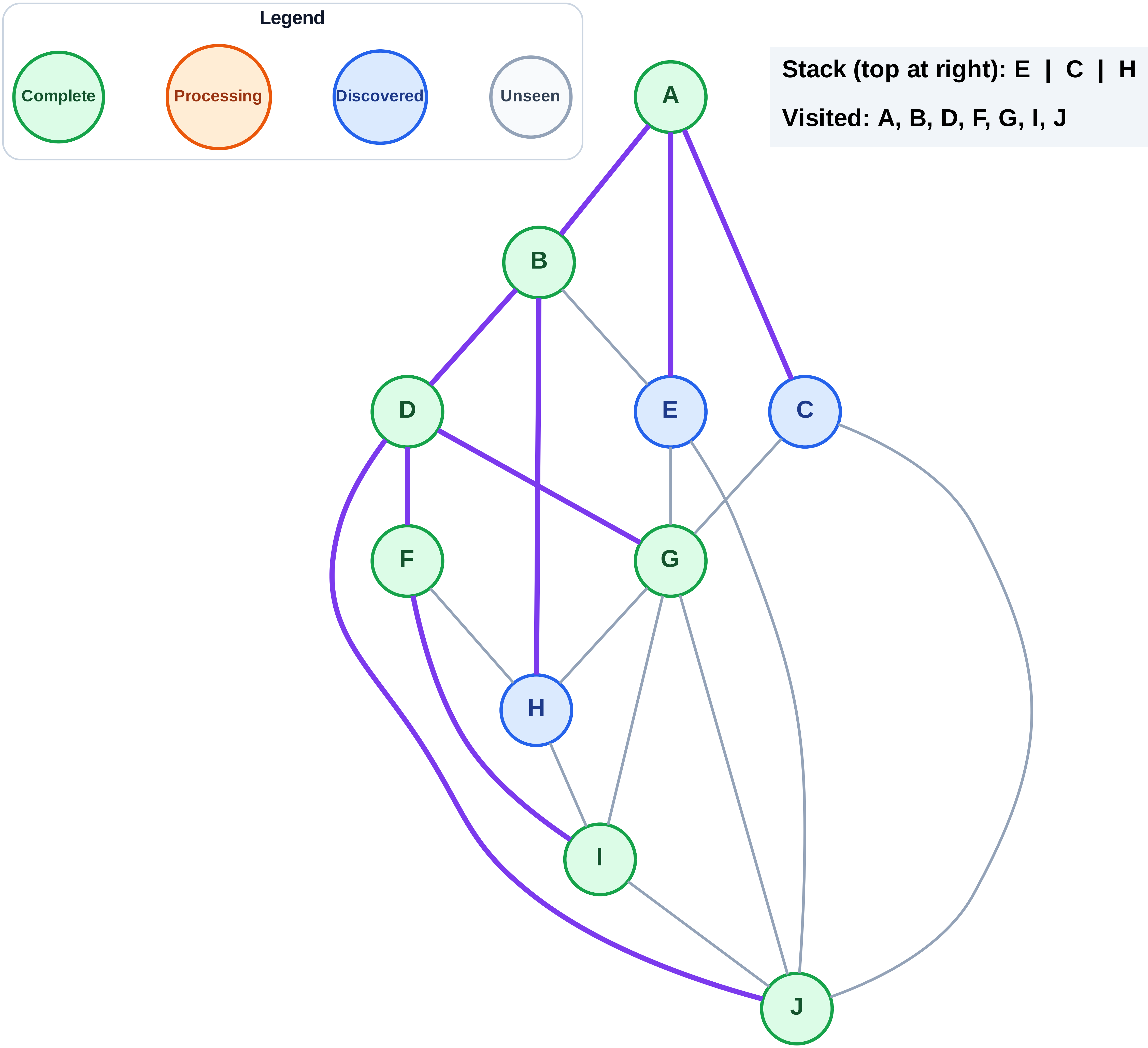
DFS Traversal

Step 14: Process node J



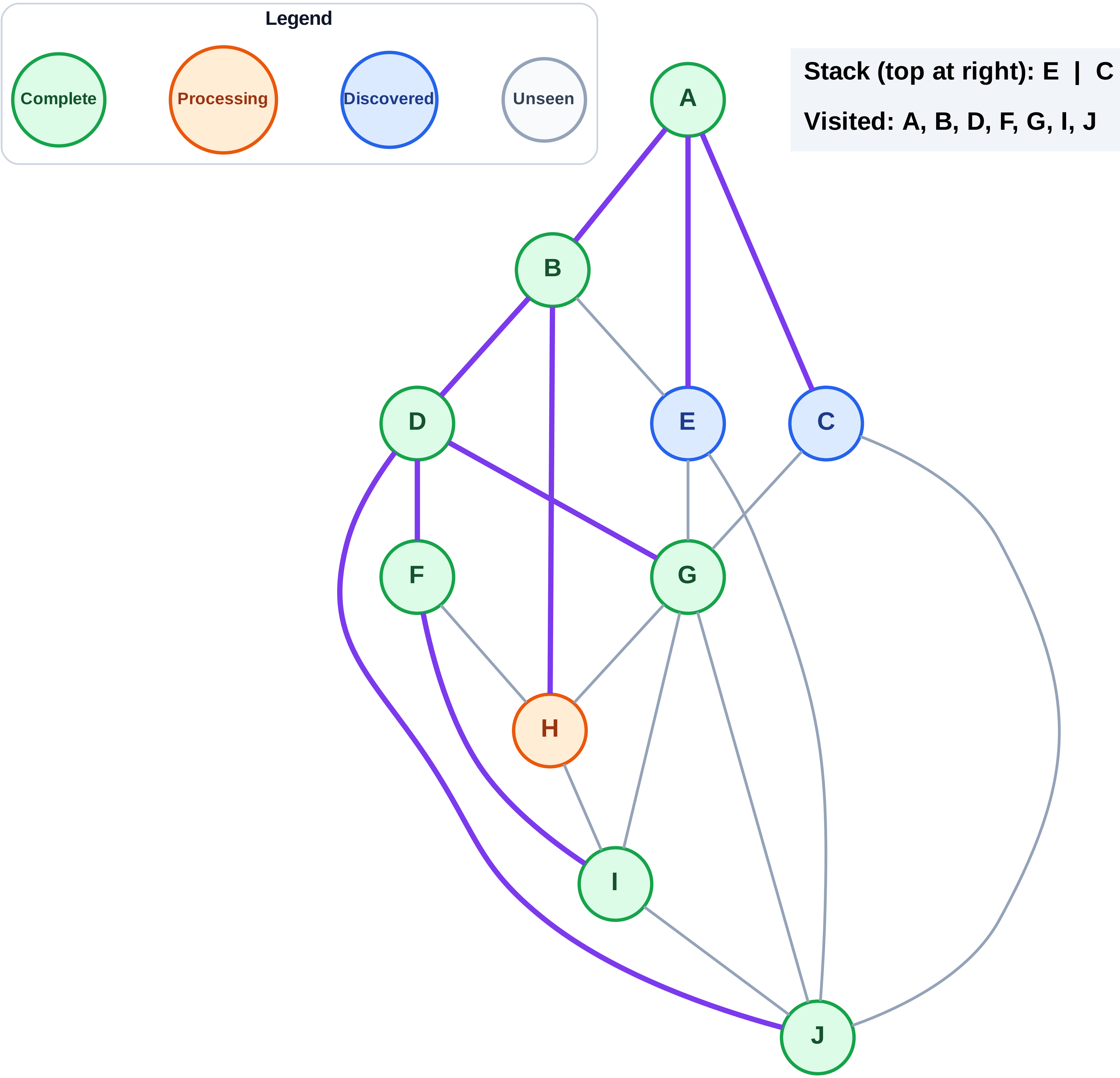
DFS Traversal

Step 15: No new neighbors from J



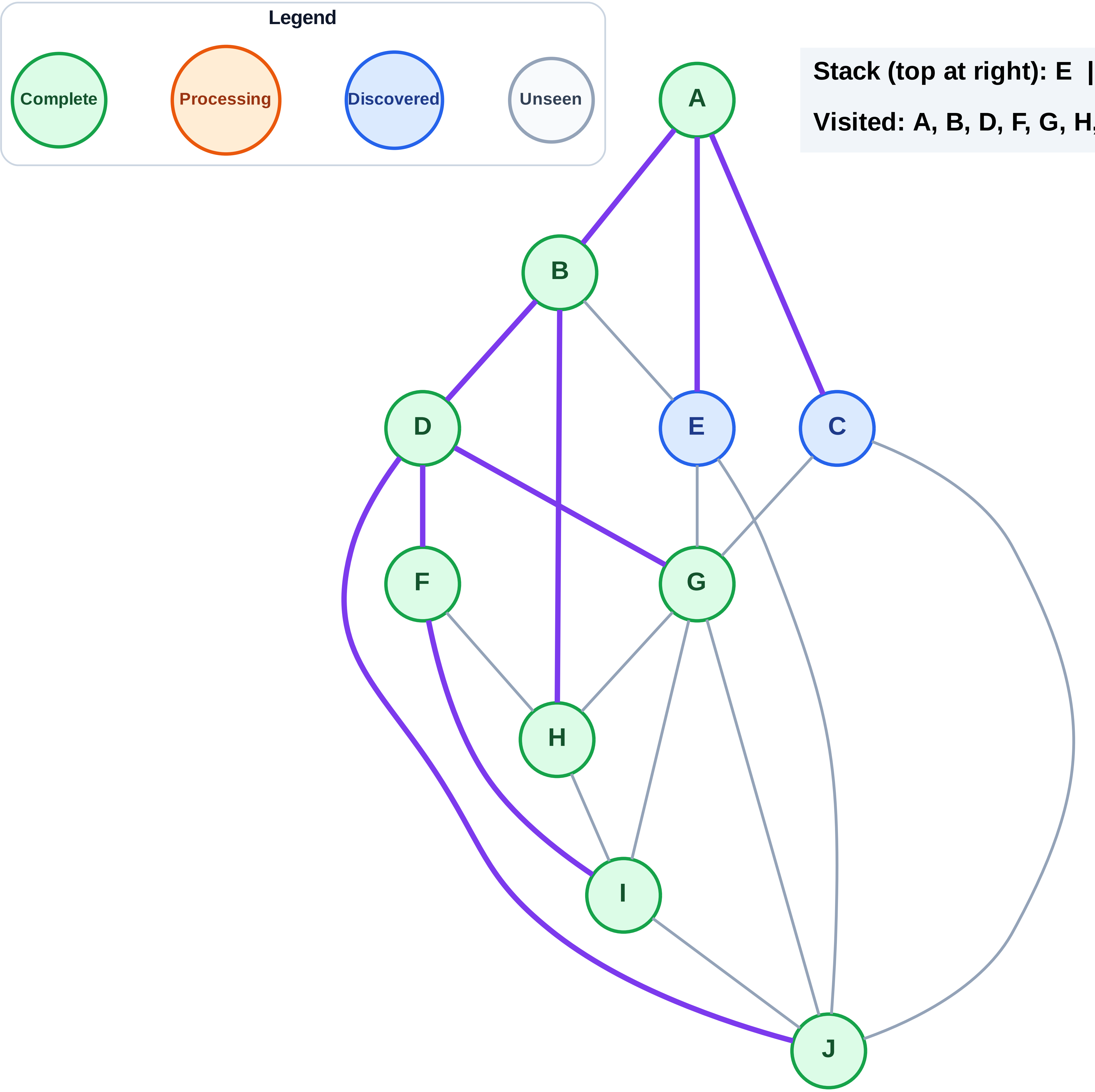
DFS Traversal

Step 16: Process node H



DFS Traversal

Step 17: No new neighbors from H

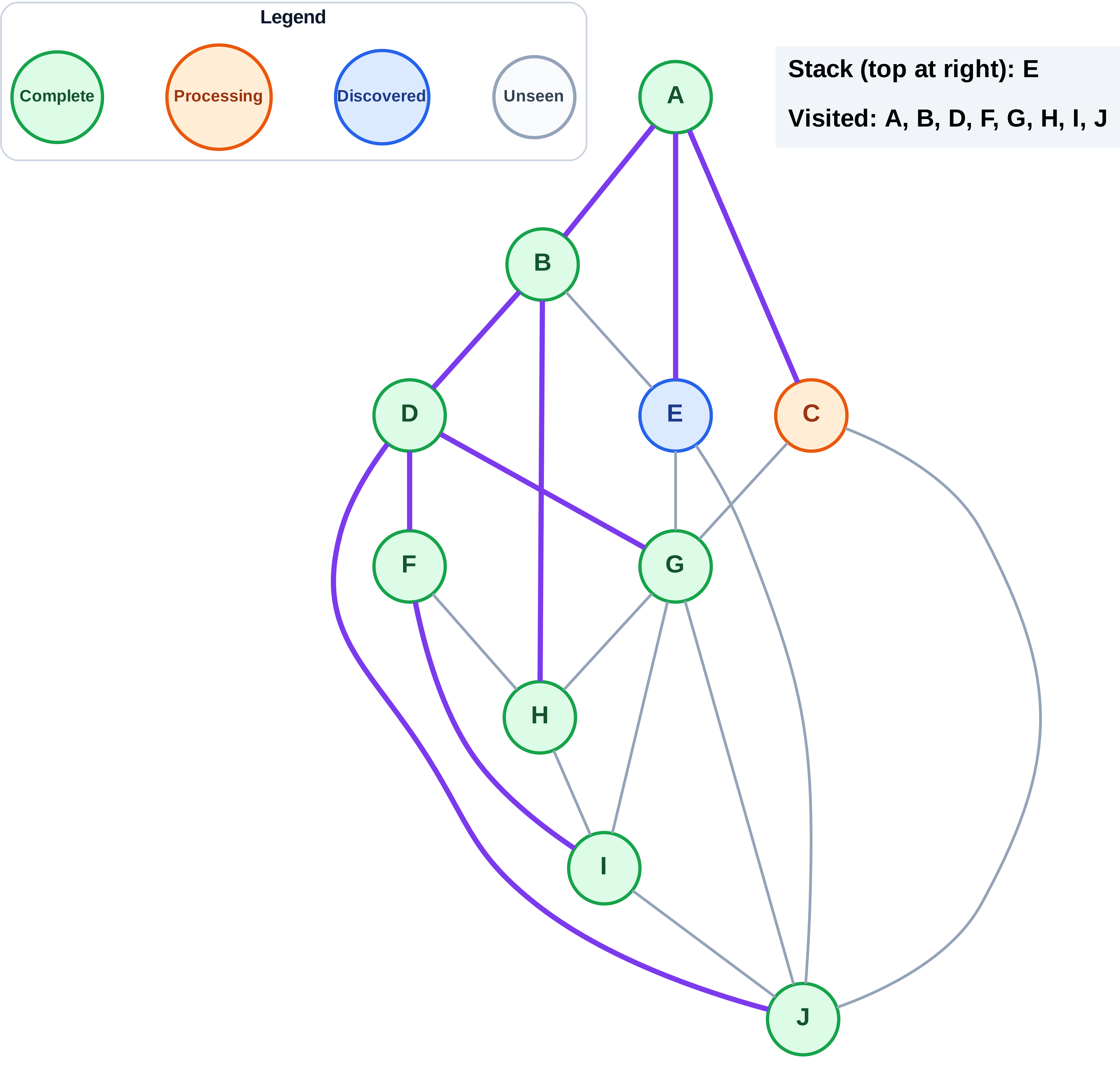


Stack (top at right): E | C

Visited: A, B, D, F, G, H, I, J

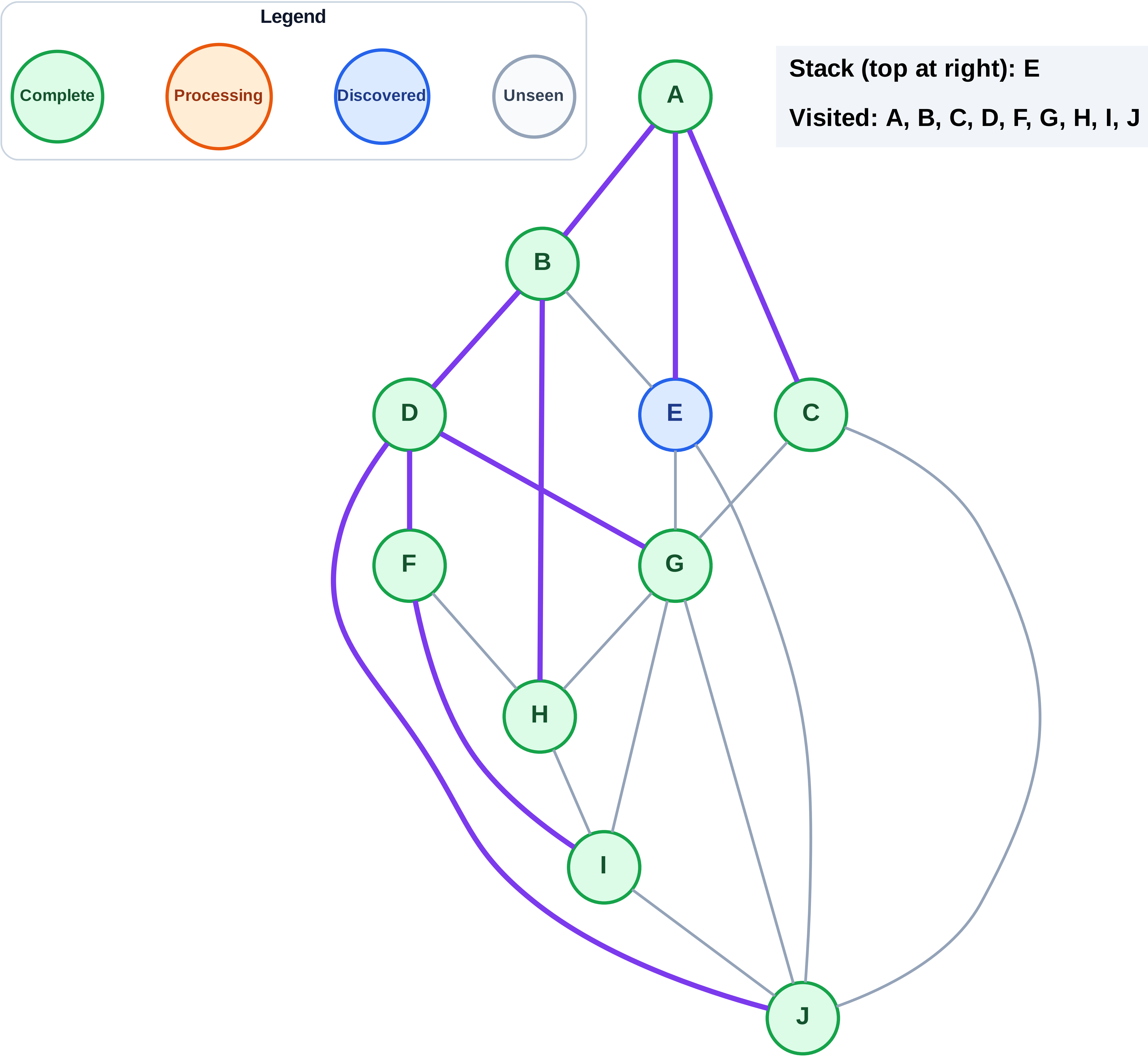
DFS Traversal

Step 18: Process node C



DFS Traversal

Step 19: No new neighbors from C

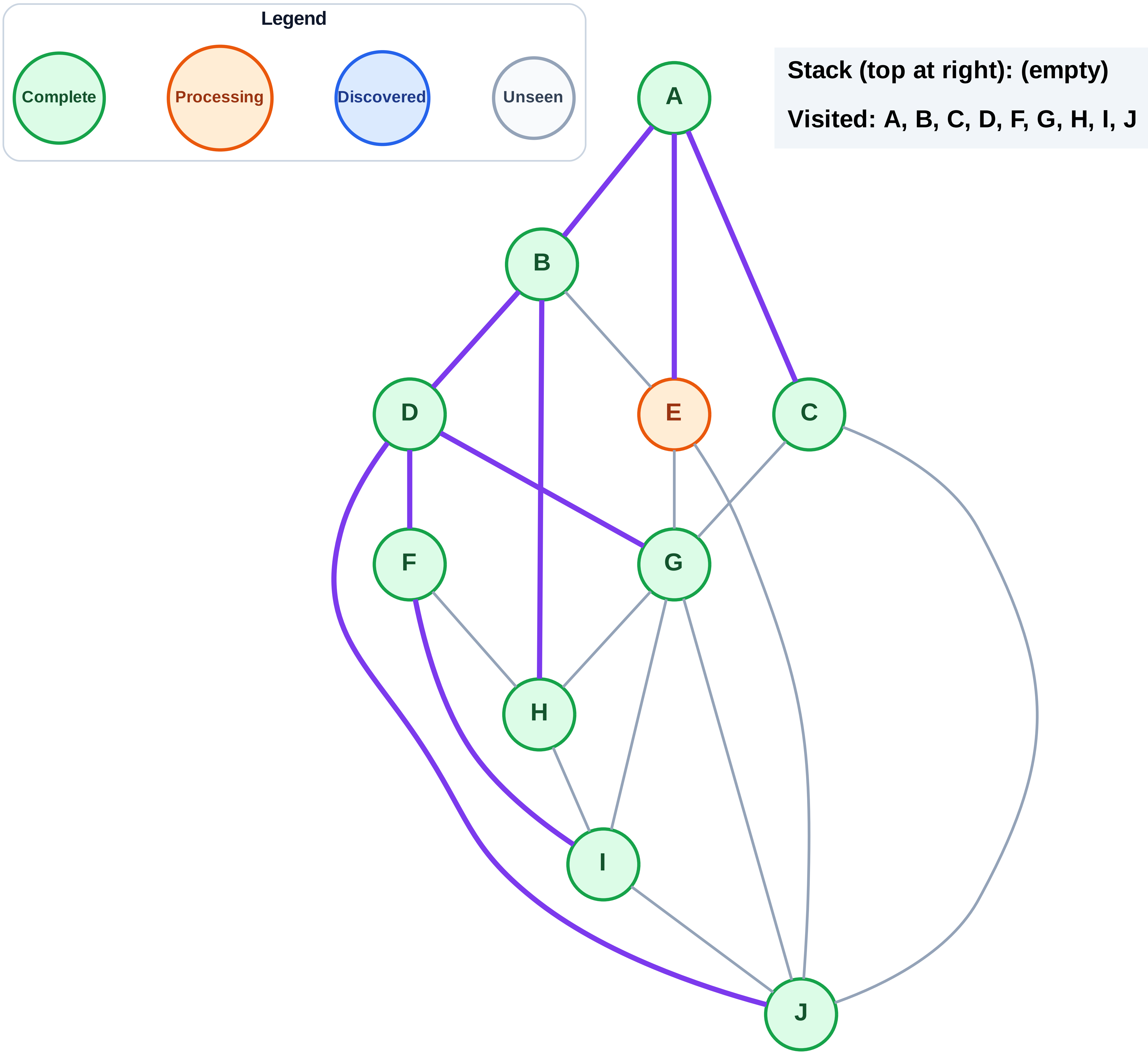


Stack (top at right): E

Visited: A, B, C, D, F, G, H, I, J

DFS Traversal

Step 20: Process node E



DFS Traversal

Step 21: No new neighbors from E